

Spark Tutorial

Introduction

In this laboratory, we will cover the basic element of programming using a map-reduce methodology. For that purpose, we will be using Apache Spark as a reference, but bear in mind that similar frameworks exists and principles can be extrapolated.

Some concepts and facts

- Spark is a distributed computing platform that operates on a cluster. Like MPI, we expect that nodes does NOT share a memory space but they are connected in high-speed dedicated network. Distributed filesystems that work over the network are extremely useful.
- It is considered the next generation of previous map-reduce standard Apache Hadoop. Main difference is thought to be the use of the memory instead of disk for intermediate operations, but there are many more improvements.
- It is built on Java. Despite this, it can be programmed using Java, Scala, Python or R. The complete API can only be found in JVM-based languages but the most frequent one is PySpark, since people is reluctant to use JVM-based languages in data science. Indeed, since Hadoop was only available for Java, it is likely that some Java codes of Spark are adaptations of previous Hadoop codes.
- Resilient Distributed Dataset (RDD): the basic unit that is processed in Spark. Equivalent to a numpy array but distributed.
- RDD API usually exposes the low-level operations of Apache Spark, useful for preprocessing data but useless for data analytics
- For data analysis, Dataframe and Spark SQL is used. It relies on a pandas-alike API that even accepts SQL code (which may sound crazy and useless for developers, but many *old* data scientists and statisticians are really proficient in SQL but not in Python).

How to install Spark

Colab already has Apache Spark and the Apache Spark client for Python (pyspark) installed.

```
In [3]: #spark_version = "3.5.8"
#hadoop_version = "3"

#!apt-get update
#!apt-get install openjdk-8-jdk-headless -qq # Install JVM v8
#!wget -O spark-{spark_version}-bin-hadoop{hadoop_version}.tgz -q https://downlo
#!tar xf spark-{spark_version}-bin-hadoop{hadoop_version}.tgz # Unzip
#import os
#os.environ["JAVA_HOME"] = "/usr/lib/jvm/java-8-openjdk-amd64"
#os.environ["SPARK_HOME"] = f"/content/spark-{spark_version}-bin-hadoop{hadoop_v

In [4]: #%pip install pyspark>=4.0.0
```

How to initialize Spark

```
In [5]: from pyspark.sql import SparkSession

APP_NAME = "CAP-lab3"
SPARK_URL = "local[*]"
spark = SparkSession.builder.appName(APP_NAME).master(SPARK_URL).getOrCreate()
sc = spark.sparkContext
```

First Part: RDDs

Basic operations

Parallelize & collect

It creates a RDD out of a list or array. Second argument indicates the number of pieces of the RDD

```
In [6]: array = sc.parallelize([1,2,3,4,5,6,7,8,9,10], 2)
array
```

Out[6]: ParallelCollectionRDD[1] at readRDDFromFile at PythonRDD.scala:297

```
In [7]: import numpy as np
randomSamples = sc.parallelize(np.random.randn(100))
randomSamples
```

Out[7]: ParallelCollectionRDD[2] at readRDDFromFile at PythonRDD.scala:297

Cool, RDDs can not be printed...

Of course, RDDs can not be printed unless they are reduced

```
In [8]: print(array.collect())
print(randomSamples.collect())
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
[np.float64(1.2190540668283312), np.float64(-0.9301560677822476), np.float64(1.0255908047841225), np.float64(-1.4828505932849507), np.float64(-0.4773474761910001), np.float64(0.7214818992651082), np.float64(0.43900049649025213), np.float64(0.784632512659628), np.float64(-0.7305971721924198), np.float64(-0.04691753928817106), np.float64(-1.3475763826271743), np.float64(-1.472978266615426), np.float64(1.750334421804924), np.float64(-0.3606614464065321), np.float64(-1.6971011031066512), np.float64(0.9715401041076726), np.float64(1.7592717088653786), np.float64(1.9544782147594841), np.float64(1.1955214140547146), np.float64(0.256541756837526), np.float64(0.6963757278080548), np.float64(1.294534362093661), np.float64(-2.0730456610049695), np.float64(-1.0498208639173081), np.float64(0.7324861605082778), np.float64(0.022067091159697266), np.float64(-0.5043214108707698), np.float64(0.6121742799140838), np.float64(0.7045553368639604), np.float64(-0.6785120936260285), np.float64(-0.8636270272847836), np.float64(-1.1181447445887893), np.float64(0.08864838197708987), np.float64(-1.4196822259330537), np.float64(1.6616184674400207), np.float64(-0.2989197392811141), np.float64(-0.5372293563116665), np.float64(-0.8577196574578272), np.float64(-0.42757801746539253), np.float64(1.0399373371176093), np.float64(0.031654234774903506), np.float64(-0.21445032196179395), np.float64(-0.40819289182192825), np.float64(1.4688871273697448), np.float64(-1.6870910436623276), np.float64(0.7354228961546687), np.float64(0.7140237511162539), np.float64(0.6898664648947725), np.float64(-0.9450671794133825), np.float64(0.8790177277943323), np.float64(0.24295898879900668), np.float64(0.9572295016002805), np.float64(-0.4114732925155852), np.float64(1.2704775553571348), np.float64(-0.8327655266778914), np.float64(1.4524204756151595), np.float64(0.4764528153127353), np.float64(1.228575340685368), np.float64(0.908368451141703), np.float64(-2.134340062533536), np.float64(1.6337699625742832), np.float64(1.362022232847681), np.float64(-1.2468090420006386), np.float64(-0.28745280804699813), np.float64(-1.142591150197565), np.float64(-1.1403740146287438), np.float64(0.3595833540369747), np.float64(-0.15748734191443042), np.float64(-0.30588467351161686), np.float64(-0.08758014123156785), np.float64(0.5768506303071709), np.float64(-0.9077146558237067), np.float64(-0.7419345161545629), np.float64(-0.2207901960017426), np.float64(1.0342398291047812), np.float64(0.934628783406928), np.float64(-0.2492440974446541), np.float64(0.5934288384910104), np.float64(-1.2155344891721473), np.float64(0.9787948425970151), np.float64(0.17461925226460792), np.float64(0.04093369607232921), np.float64(1.0063995691203647), np.float64(0.23151366074913085), np.float64(-0.058552234735244894), np.float64(-1.347830541795793), np.float64(-0.8273353452151625), np.float64(0.9549332236966485), np.float64(1.144436096773951), np.float64(-0.04591899259042625), np.float64(-0.6741589224650691), np.float64(-0.056070246621611246), np.float64(-0.26456413708858484), np.float64(-0.9569832607128531), np.float64(-0.22376754427966167), np.float64(0.21148928359305588), np.float64(-0.10867907370442213), np.float64(-0.006917826349141942), np.float64(-1.8042935801949025), np.float64(-0.4691816484546272)]
```

Spark uses lazy operations for everything, this means that nothing is evaluated until an action, a reduce operation normally, is performed. The basic reduce operation is collect, which returns the whole RDD (i.e. no reduction is performed).

Other ways of loading data

In [9]: `import requests`

```
request = requests.get("https://gist.githubusercontent.com/jsdario/6d6c69398cb0c")
with open("elquijote.txt", "wb") as f:
    f.write(request.content)
```

In [10]: `quijote = sc.textFile("elquijote.txt")`
`quijote.take(10)`

```
Out[10]: ['DON QUIJOTE DE LA MANCHA',
'Miguel de Cervantes Saavedra',
'',
'PRIMERA PARTE',
'CAPÍTULO 1: Que trata de la condición y ejercicio del famoso hidalgo D. Quijote de la Mancha',
'En un lugar de la Mancha, de cuyo nombre no quiero acordarme, no ha mucho tiempo que vivía un hidalgo de los de lanza en astillero, adarga antigua, rocín flaco y galgo corredor. Una olla de algo más vaca que carnero, salpicón las más noches, duelos y quebrantos los sábados, lentejas los viernes, algún palomino de añadidura los domingos, consumían las tres partes de su hacienda. El resto della concluían sayo de velarte, calzas de velludo para las fiestas con sus pantufleros de lo mismo, los días de entre semana se honraba con su vellori de lo más fino. Tenía en su casa una ama que pasaba de los cuarenta, y una sobrina que no llegaba a los veinte, y un mozo de campo y plaza, que así ensillaba el rocín como tomaba la podadera. Frisaba la edad de nuestro hidalgo con los cincuenta años, era de complexión recia, seco de carnes, enjuto de rostro; gran madrugador y amigo de la caza. Quieren decir que tenía el sobrenombre de Quijada o Quesada (que en esto hay alguna diferencia en los autores que deste caso escriben), aunque por conjeturas verosímiles se deja entender que se llama Quijana; pero esto importa poco a nuestro cuento; basta que en la narración dél no se salga un punto de la verdad. Es, pues, de saber, que este sobredicho hidalgo, los ratos que estaba ocioso (que eran los más del año) se daba a leer libros de caballerías con tanta afición y gusto, que olvidó casi de todo punto el ejercicio de la caza, y aun la administración de su hacienda; y llegó a tanto su curiosidad y desatino en esto, que vendió muchas hanegas de tierra de sembradura, para comprar libros de caballerías en que leer; y así llevó a su casa todos cuantos pudo haber dellos; y de todos ningunos le parecían tan bien como los que compuso el famoso Feliciano de Silva: porque la claridad de su prosa, y aquellas intrincadas razones suyas, le parecían de perlas; y más cuando llegaba a leer aquellos requiebros y cartas de desafío, donde en muchas partes hallaba escrito: la razón de la sinrazón que a mi razón se hace, de tal manera mi razón enflaquece, que con razón me quejo de la vuestra fermosura, y también cuando leía: los altos cielos que de vuestra divinidad divinamente con las estrellas se fortifican, y os hacen merecedora del merecimiento que merece la vuestra grandeza. Con estas y semejantes razones perdía el pobre caballero el juicio, y desvelábase por entenderlas, y desentrañarles el sentido, que no se lo sacara, ni las entendiera el mismo Aristóteles, si resucitara para sólo ello. No estaba muy bien con las heridas que don Belianis daba y recibía, porque se imaginaba que por grandes maestros que le hubiesen curado, no dejaría de tener el rostro y todo el cuerpo lleno de cicatrices y señales; pero con todo alababa en su autor aquel acabar su libro con la promesa de aquella inacabable aventura, y muchas veces le vino deseo de tomar la pluma, y darle fin al pie de la letra como allí se promete; y sin duda alguna lo hiciera, y aun saliera con ello, si otros mayores y continuos pensamientos no se lo estorbaran.',
'Tuvo muchas veces competencia con el cura de su lugar (que era hombre docto graduado en Sigüenza), sobre cuál había sido mejor caballero, Palmerín de Inglaterra o Amadís de Gaula; mas maese Nicolás, barbero del mismo pueblo, decía que ninguno llegaba al caballero del Febo, y que si alguno se le podía comparar, era don Galaor, hermano de Amadís de Gaula, porque tenía muy acomodada condición para todo; que no era caballero melindroso, ni tan llorón como su hermano, y que en lo de la valentía no le iba en zaga.',
'En resolución, él se enfrascó tanto en su lectura, que se le pasaban las noches leyendo de claro en claro, y los días de turbio en turbio, y así, del poco dormir y del mucho leer, se le secó el cerebro, de manera que vino a perder el juicio. Llenósele la fantasía de todo aquello que leía en los libros, así de encantamientos, como de pendencias, batallas, desafíos, heridas, requiebros, amores, tormentas y disparates imposibles, y asentósele de tal modo en la imaginación que era verdad toda aquella máquina de aquellas soñadas invenciones que leía, que para él no había otra',
```

```
'historia más cierta en el mundo.',
'Decía él, que el Cid Ruy Díaz había sido muy buen caballero; pero que no tenía que ver con el caballero de la ardiente espada, que de sólo un revés había partido por medio dos fieros y descomunales gigantes. Mejor estaba con Bernardo del Carpio, porque en Roncesvalle había muerto a Roldán el encantado, valiéndose de la industria de Hércules, cuando ahogó a Anteo, el hijo de la Tierra, entre los brazos. Decía mucho bien del gigante Morgante, porque con ser de aquella generación gigantesca, que todos son soberbios y descomedidos, él solo era afable y bien criado; pero sobre todos estaba bien con Reinaldos de Montalbán, y más cuando le veía salir de su castillo y robar cuantos topaba, y cuando en Alende robó aquel ídolo de Mahoma, que era todo de oro, según dice su historia. Diera él, por dar una mano de coces al traidor de Galalón, al ama que tenía y aun a su sobrina de añadidura.']
```

Here, you can see both a method to load a text file line per line and a another reduction operation.

```
In [11]: quijote.take?
```

Transformations

Let's review all the transformation that can be performed to data.

```
In [12]: charsPerLine = quijote.map(lambda s: len(s))
allWords = quijote.flatMap(lambda s: s.split())
allWordsNoArticles = allWords.filter(lambda a: a.lower() not in ["el", "la"])
allWordsUnique = allWords.map(lambda s: s.lower()).distinct()
sampleWords = allWords.sample(withReplacement=True, fraction=0.2, seed=666)
weirdSampling = sampleWords.union(allWordsNoArticles.sample(False, fraction=0.3))
```

```
In [13]: allWordsUnique.take(10)
```

```
Out[13]: ['don',
'quijote',
'la',
'miguel',
'cervantes',
'1:',
'condición',
'ejercicio',
'famoso',
'hidalgo']
```

Assignment question

Explain the use and purpose of each operation above.

Comment also on the size of the resulting RDD in terms of the size of the original RDD, e.g. if original RDD is of size N , then `rdd.filter()` is of size $K \leq N$

Answer:

- map: Aplica una función a cada elemento de una RDD
 - flatMap: Aplica una función a cada elemento y luego aplana o combina los resultados
 - filter: Filtra elementos según una condición
 - distinct Elimina duplicados
 - sample: Toma una muestra aleatoria del RDD
 - union: Une dos RDDs
-

Actions

```
In [14]: numLines = quijote.count()
numChars = charsPerLine.reduce(lambda a,b: a+b) # also charsPerLine.sum()
sortedWordsByLength = allWordsNoArticles.takeOrdered(10, key=lambda x: -len(x))
numLines, numChars, sortedWordsByLength
```

```
Out[14]: (2186,
1036211,
['procuremos.Levántate,',
'extraordinariamente,',
'estrechísimamente,',
'convirtiéndoseles',
'entretenimientos,',
'inadvertidamente.',
'cortesísimamente',
'Agredeciéronselo',
'Pintiquiniestra,',
'entretenimiento,'])
```

Assignment question

Explain the use and purpose of each action above.

- Line 1: Cuenta el número total de elementos del RDD
- Line 2: Agrega todos los elementos del RDD usando una función
- Line 3: Devuelve los 10 elementos "mínimos" según una clave (aquí, longitud negativa)

Implement the count operation using reduce as the unique option. You can use transformations. Is it possible to achieve a solution without any transformation? Does it make sense?

Answer: count = quijote.map(lambda x: 1).reduce(lambda a, b: a + b)

No se puede hacer sin transformaciones, porque necesitas valores que se puedan agregar. Si no lo haces así romperías el paralelismo.

Key-Value RDDs

```
In [15]: import re
allWords = allWords.flatMap(lambda w: re.sub("";|:|\.|,|-|-|'|'\s""", " ", w.lower()).split(" ")).filter(lambda a: len(a)>0)
allWords2 = sc.parallelize(requests.get("https://gist.githubusercontent.com/jsda
allWords2 = allWords2.flatMap(lambda w: re.sub("";|:|\.|,|-|-|'|'\s""", " ", w.
```

```
<>:2: SyntaxWarning: invalid escape sequence '\.'
<>:4: SyntaxWarning: invalid escape sequence '\.'
<>:2: SyntaxWarning: invalid escape sequence '\.'
<>:4: SyntaxWarning: invalid escape sequence '\.'
/tmp/ipykernel_1006/455926106.py:2: SyntaxWarning: invalid escape sequence '\.'
    allWords = allWords.flatMap(lambda w: re.sub("";|:|\.|,|-|-|'|'\s""", " ", w.lower()).split(" ")).filter(lambda a: len(a)>0)
/tmp/ipykernel_1006/455926106.py:4: SyntaxWarning: invalid escape sequence '\.'
    allWords2 = allWords2.flatMap(lambda w: re.sub("";|:|\.|,|-|-|'|'\s""", " ", w.decode("utf8").lower()).split(" ")).filter(lambda a: len(a)>0)
```

```
In [16]: allWords.take(10), allWords2.take(10)
```

```
Out[16]: (['don',
            'quijote',
            'de',
            'la',
            'mancha',
            'miguel',
            'de',
            'cervantes',
            'saavedra',
            'primera'],
          ['don',
            'quijote',
            'de',
            'la',
            'mancha',
            'miguel',
            'de',
            'cervantes',
            'saavedra',
            'segunda'])
```

Next, we move to more interesting operations that involve key-value RDDs. Key-value RDDs are a special kind of RDDs where each element is a tuple (K,V) where K is the key and V the value.

```
In [17]: words = allWords.map(lambda e: (e,1))
words2 = allWords2.map(lambda e: (e,1))

words.take(10)
```

```
Out[17]: [('don', 1),
          ('quijote', 1),
          ('de', 1),
          ('la', 1),
          ('mancha', 1),
          ('miguel', 1),
          ('de', 1),
          ('cervantes', 1),
          ('saavedra', 1),
          ('primera', 1)]
```

How to manipulate K-V RDDs

```
In [18]: frequencies = words.reduceByKey(lambda a,b: a+b)
         frequencies2 = words2.reduceByKey(lambda a,b: a+b)
         frequencies.takeOrdered(10, key=lambda a: -a[1])
```

```
Out[18]: [('que', 10705),
          ('de', 9033),
          ('y', 8668),
          ('la', 5015),
          ('a', 4815),
          ('en', 4046),
          ('el', 3857),
          ('no', 3083),
          ('se', 2382),
          ('los', 2148)]
```

```
In [19]: res = words.groupByKey().takeOrdered(10, key=lambda a: -len(a))
         res # To see the content, res[i][1].data
```

```
Out[19]: [('don', <pyspark.resultiterable.ResultIterable at 0x7ec184e0eb40>),
          ('quijote', <pyspark.resultiterable.ResultIterable at 0x7ec184e0e690>),
          ('la', <pyspark.resultiterable.ResultIterable at 0x7ec184e0e720>),
          ('miguel', <pyspark.resultiterable.ResultIterable at 0x7ec184e0eab0>),
          ('cervantes', <pyspark.resultiterable.ResultIterable at 0x7ec184e0ee70>),
          ('condición', <pyspark.resultiterable.ResultIterable at 0x7ec184e0ef60>),
          ('ejercicio', <pyspark.resultiterable.ResultIterable at 0x7ec184e0eff0>),
          ('famoso', <pyspark.resultiterable.ResultIterable at 0x7ec184e0f0b0>),
          ('hidalgo', <pyspark.resultiterable.ResultIterable at 0x7ec184e0f020>),
          ('d', <pyspark.resultiterable.ResultIterable at 0x7ec184e0f170>)]
```

```
In [20]: joinFreq = frequencies.join(frequencies2)
         joinFreq.take(10)
```

```
Out[20]: [('don', (1072, 1606)),
          ('quijote', (846, 1354)),
          ('la', (5015, 5461)),
          ('miguel', (3, 5)),
          ('cervantes', (3, 1)),
          ('condición', (33, 39)),
          ('ejercicio', (22, 31)),
          ('famoso', (52, 37)),
          ('hidalgo', (22, 42)),
          ('un', (900, 1062))]
```

```
In [21]: joinFreq.map(lambda e: (e[0], (e[1][0] - e[1][1])/(e[1][0] + e[1][1]))).takeOrde
```



```
Out[21]: ([('bacía', 0.9393939393939394),
('venia', 0.9230769230769231),
('hermandad', 0.9),
('andrés', 0.8823529411764706),
('micomicona', 0.8823529411764706),
('peña', 0.8823529411764706),
('barca', 0.875),
('yerme', 0.875),
('novela', 0.875),
('acertó', 0.8666666666666667)],
[('teresa', -0.9767441860465116),
('roque', -0.96),
('refranes', -0.9375),
('condesa', -0.9333333333333333),
('leones', -0.9333333333333333),
('gobernadores', -0.9166666666666666),
('lacayo', -0.9166666666666666),
('visorrey', -0.9130434782608695),
('antonio', -0.9076923076923077),
('zaragoza', -0.9047619047619048)])
```

Assignment question

Explain the use and purpose of each action above.

- `reduceByKey`: Agrega valores por clave (suma de frecuencias), calcular conteo de palabras.
- `takeOrdered`: Obtener top 10 palabras más frecuentes
- `join`: Une dos RDDs por clave en concreto `frequencies` y `frequencies2`

Implement the frequency with `groupByKey` and transformations. `frequencies = words.reduceByKey(lambda a, b: a + b)`

Para ver las 10 palabras más comunes:

```
sorted_frequencies = frequencies.takeOrdered(10, key=lambda x: -x[1])
```

Which of the two following cells is more efficient?

Answer: La segunda celda es más eficiente (y mejor práctica de programación) porque define el grafo de computación una sola vez. En la primera celda, Spark tiene que parsear y planificar la misma transformación dos veces por separado.

```
In [52]: joinFreq.map(lambda e: (e[0], (e[1][0] - e[1][1])/(e[1][0] + e[1][1]))).takeOrdered
```

```
Out[52]: ([('bacía', 0.9393939393939394),
('venia', 0.9230769230769231),
('hermandad', 0.9),
('andr s', 0.8823529411764706),
('micomicona', 0.8823529411764706),
('pe a', 0.8823529411764706),
('barca', 0.875),
('yerme', 0.875),
('novela', 0.875),
('acert ', 0.8666666666666667)],
[('teresa', -0.9767441860465116),
('roque', -0.96),
('refranes', -0.9375),
('condesa', -0.9333333333333333),
('leones', -0.9333333333333333),
('gobernadores', -0.9166666666666666),
('lacayo', -0.9166666666666666),
('visorrey', -0.9130434782608695),
('antonio', -0.9076923076923077),
('zaragoza', -0.9047619047619048)])
```

```
In [53]: result = joinFreq.map(lambda e: (e[0], (e[1][0] - e[1][1])/(e[1][0] + e[1][1])))
result.takeOrdered(10, lambda v: -v[1]), result.takeOrdered(10, lambda v: +v[1])
```

```
Out[53]: ([('bacía', 0.9393939393939394),
('venia', 0.9230769230769231),
('hermandad', 0.9),
('andr s', 0.8823529411764706),
('micomicona', 0.8823529411764706),
('pe a', 0.8823529411764706),
('barca', 0.875),
('yerme', 0.875),
('novela', 0.875),
('acert ', 0.8666666666666667)],
[('teresa', -0.9767441860465116),
('roque', -0.96),
('refranes', -0.9375),
('condesa', -0.9333333333333333),
('leones', -0.9333333333333333),
('gobernadores', -0.9166666666666666),
('lacayo', -0.9166666666666666),
('visorrey', -0.9130434782608695),
('antonio', -0.9076923076923077),
('zaragoza', -0.9047619047619048)])
```

Optimizations and final notes

Optimizing the data movement around the cluster

One of the main issues could be that if data after an operation is not balanced, we may not be using the cluster properly. For that purpose, we have two operations

```
In [24]: result.coalesce(numPartitions=2) # Avoids the data movement, so it tries to bala
result.repartition(numPartitions=2) # We don't care about data movement, this ba
```

```
Out[24]: MapPartitionsRDD[51] at coalesce at NativeMethodAccessorImpl.java:0
```

Persistence for intermediate operations

In contrast to Hadoop, intermediate RDDs are not preserved, each time we use an action/reduction, the whole data pipeline is executed from the datasources. To avoid this:

```
In [25]: result.take(10)
allWords.cache() # allWords RDD must stay in memory after computation, we made
result.take(10)
```

```
Out[25]: [('don', -0.19940253920836445),
('quijote', -0.2309090909090909),
('la', -0.04257350133638793),
('miguel', -0.25),
('cervantes', 0.5),
('condición', -0.08333333333333333),
('ejercicio', -0.16981132075471697),
('famoso', 0.16853932584269662),
('hidalgo', -0.3125),
('un', -0.08256880733944955)]
```

```
In [26]: from pyspark import StorageLevel
# https://spark.apache.org/docs/2.2.0/rdd-programming-guide.html#rdd-persistence
allWords2.persist(StorageLevel.MEMORY_AND_DISK) # Now it will be preserved on di
```

```
Out[26]: PythonRDD[55] at RDD at PythonRDD.scala:56
```

```
In [59]: !rm -rf palabras_parte2
allWords2.coalesce(numPartitions=5)
allWords2.saveAsTextFile("palabras_parte2")
```

```
In [60]: !ls palabras_parte2

part-00000 part-00001 _SUCCESS
```

Assignment question

Before saving with saveAsTextFile, use coalesce with different values. What's the difference in the previous ls?

Answer: Vemos que segun el número de particiones que creamos se harán más o menos archivos.

Global variables

There are two kind of global variables, read-only and write-only.

```
In [29]: articles = sc.broadcast(["el", "la"])
articles.value
```

```
Out[29]: ['el', 'la']
```

Broadcast variables are read-only. They help us to avoid local variables of the closures (the functions we use inside map, reduce, ...) to be transferred in every single Spark operation. In that way, they are only transferred only once.

```
In [30]: acc = sc.accumulator(0)
def incrementar(x):
    global acc
    acc += x

allWords.map(lambda l:1).foreach(incrementar)
acc
```

```
Out[30]: Accumulator<id=0, value=187045>
```

Write-only variables can be also declared and initialized, but they can not be read since reading will force a complete synchronization of the cluster.

Second part: Spark SQL

Next, we will do a short review of the high-level API of Spark

```
In [31]: !wget --no-check-certificate 'https://docs.google.com/uc?export=download&id=1D1Cj
--2026-04-13 15:20:49-- https://docs.google.com/uc?export=download&id=1D1CjjqZtv
upN8_K1s4dBYUP8PQUPFIVl
Resolving docs.google.com (docs.google.com)... 142.251.163.101, 142.251.163.139,
142.251.163.102, ...
Connecting to docs.google.com (docs.google.com)|142.251.163.101|:443... connecte
d.
HTTP request sent, awaiting response... 303 See Other
Location: https://drive.usercontent.google.com/download?id=1D1CjjqZtvupN8_K1s4dBY
UP8PQUPFIVl&export=download [following]
--2026-04-13 15:20:49-- https://drive.usercontent.google.com/download?id=1D1Cjjq
ZtvupN8_K1s4dBYUP8PQUPFIVl&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 192.178.
155.132, 2607:f8b0:4004:c1d::84
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)|192.17
8.155.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 37192870 (35M) [application/octet-stream]
Saving to: 'reddit_02_2016.parquet'

reddit_02_2016.parq 100%[=====>] 35.47M 44.2MB/s in 0.8s

2026-04-13 15:20:52 (44.2 MB/s) - 'reddit_02_2016.parquet' saved [37192870/371928
70]
```

```
In [32]: import pandas as pd
```

```
df = spark.read.parquet("reddit_02_2016.parquet")
```

```
In [33]: df.show()
```

```
+-----+-----+-----+-----+-----+
|author|created_utc|title|subreddit|
selftext|over_18|
+-----+-----+-----+-----+
|BlairTheWiseViking| 1454326164|(24M) with a DB p...|deadbedroom|This po
st will be...| false|
|DAVIDSPZGZ| 1454326165|EAL: Rebajas Sony...|ElAndroideLibre|
| false|
|chasdave1981| 1454326165| Ms Juicy (retired)|bigasses|
| true|
|[deleted]| 1454326165|[H] ST MW Aquamar...|GlobalOffensiveTrade|
[deleted]| true|
|SableArgyle| 1454326166|Something might h...|OnePiece|So waaa
aaaaay bac...| true|
|nilimajaveri| 1454326166|Buy queen bed mat...|InternetIsBeautiful|
| false|
|[deleted]| 1454326166|Grand entrance|gifs|
[deleted]| false|
|a2zlifestyles| 1454326167|9 Cool Indian Hai...|HairStyle|
| false|
|dingdong89| 1454326169|Robert Leslie - I...|BinauralMusic|
| false|
|sleepyechoes| 1454326170|Presidential Foot...|CFB|I put t
ogether th...| false|
|rotoreuters| 1454326171|David Frost son d...|betternews|
| false|
|rotoreuters| 1454326171|Syria: former UK ...|betternews|
| false|
|rotoreuters| 1454326171|The forgotten sto...|betternews|
| false|
|rotoreuters| 1454326172|British woman con...|betternews|
| false|
|kaizoku99| 1454326172|Income sharing ag...|studyAbroad|
[removed]| false|
|rotoreuters| 1454326173|Russian Foreign M...|betternews|
| false|
|Tectonic_| 1454326173|The most depressi...|GlobalOffensive|
| false|
|rotoreuters| 1454326173|Russian lawmaker ...|betternews|
| false|
|rotoreuters| 1454326174|Australia's only ...|betternews|
| false|
|rotoreuters| 1454326174|Thor 3: Ragnarok ...|betternews|
| false|
+-----+-----+-----+-----+
only showing top 20 rows
```

Basic operations

Filtering

```
In [34]: df.filter(~df.over_18).show()
```

```
+-----+-----+-----+-----+-----+
|          author|created_utc|          title|          subreddit|
selftext|over_18|
+-----+-----+-----+-----+-----+
|BlairTheWiseViking| 1454326164|(24M) with a DB p...|          deadbedroom|This pos
t will be...| false|
|          DAVIDSPZGZ| 1454326165|EAL: Rebajas Sony...|          ElAndroideLibre|
| false|
|          nilimajaveri| 1454326166|Buy queen bed mat...|InternetIsBeautiful|
| false|
|          [deleted]| 1454326166|          Grand entrance|          gifs|
[deleted]| false|
|          a2zlifestyles| 1454326167|9 Cool Indian Hai...|          HairStyle|
| false|
|          dingdong89| 1454326169|Robert Leslie - I...|          BinauralMusic|
| false|
|          sleepyechoes| 1454326170|Presidential Foot...|          CFB|I put to
gether th...| false|
|          roforeuters| 1454326171|David Frost son d...|          betternews|
| false|
|          roforeuters| 1454326171|Syria: former UK ...|          betternews|
| false|
|          roforeuters| 1454326171|The forgotten sto...|          betternews|
| false|
|          roforeuters| 1454326172|British woman con...|          betternews|
| false|
|          kaizoku99| 1454326172|Income sharing ag...|          studyAbroad|
[removed]| false|
|          roforeuters| 1454326173|Russian Foreign M...|          betternews|
| false|
|          Tectonic_| 1454326173|The most depressi...|          GlobalOffensive|
| false|
|          roforeuters| 1454326173|Russian lawmaker ...|          betternews|
| false|
|          roforeuters| 1454326174|Australia's only ...|          betternews|
| false|
|          roforeuters| 1454326174|Thor 3: Ragnarok ...|          betternews|
| false|
|          roforeuters| 1454326174|Ofcom warns 02 ta...|          betternews|
| false|
|          roforeuters| 1454326174|Indian police hun...|          betternews|
| false|
|          jumanji112| 1454326174|The need for psyc...|          Social_Psychology|
[removed]| false|
+-----+-----+-----+-----+-----+
only showing top 20 rows
```

```
In [35]: df.where(~df.over_18).show()
```

```

+-----+-----+-----+-----+-----+
+-----+-----+
|          author|created_utc|          title|          subreddit|
selftext|over_18|
+-----+-----+-----+-----+-----+
+-----+-----+
|BlairTheWiseViking| 1454326164|(24M) with a DB p...|          deadbedroom|This pos
t will be...| false|
|          DAVIDSPZGZ| 1454326165|EAL: Rebajas Sony...|          ElAndroideLibre|
| false|
|          nilimajaveri| 1454326166|Buy queen bed mat...|InternetIsBeautiful|
| false|
|          [deleted]| 1454326166|          Grand entrance|          gifs|
[deleted]| false|
|          a2zlifestyles| 1454326167|9 Cool Indian Hai...|          HairStyle|
| false|
|          dingdong89| 1454326169|Robert Leslie - I...|          BinauralMusic|
| false|
|          sleepyechoes| 1454326170|Presidential Foot...|          CFB|I put to
gether th...| false|
|          rothereuters| 1454326171|David Frost son d...|          betternews|
| false|
|          rothereuters| 1454326171|Syria: former UK ...|          betternews|
| false|
|          rothereuters| 1454326171|The forgotten sto...|          betternews|
| false|
|          rothereuters| 1454326172|British woman con...|          betternews|
| false|
|          kaizoku99| 1454326172|Income sharing ag...|          studyAbroad|
[removed]| false|
|          rothereuters| 1454326173|Russian Foreign M...|          betternews|
| false|
|          Tectonic_| 1454326173|The most depressi...|          GlobalOffensive|
| false|
|          rothereuters| 1454326173|Russian lawmaker ...|          betternews|
| false|
|          rothereuters| 1454326174|Australia's only ...|          betternews|
| false|
|          rothereuters| 1454326174|Thor 3: Ragnarok ...|          betternews|
| false|
|          rothereuters| 1454326174|Ofcom warns O2 ta...|          betternews|
| false|
|          rothereuters| 1454326174|Indian police hun...|          betternews|
| false|
|          jumanji112| 1454326174|The need for psyc...|          Social_Psychology|
[removed]| false|
+-----+-----+-----+-----+-----+
+-----+-----+
only showing top 20 rows

```

```
In [36]: df.where("not over_18").show() # SQL syntax
```

```

+-----+-----+-----+-----+-----+
-----+-----+
|          author|created_utc|          title|          subreddit|
selftext|over_18|
+-----+-----+-----+-----+-----+
-----+-----+
|BlairTheWiseViking| 1454326164|(24M) with a DB p...|          deadbedroom|This pos
t will be...| false|
|          DAVIDSPZGZ| 1454326165|EAL: Rebajas Sony...|          ElAndroideLibre|
| false|
|          nilimajaveri| 1454326166|Buy queen bed mat...|InternetIsBeautiful|
| false|
|          [deleted]| 1454326166|          Grand entrance|          gifs|
[deleted]| false|
|          a2zllifestyles| 1454326167|9 Cool Indian Hai...|          HairStyle|
| false|
|          dingdong89| 1454326169|Robert Leslie - I...|          BinauralMusic|
| false|
|          sleepyechoes| 1454326170|Presidential Foot...|          CFB|I put to
gether th...| false|
|          rotoreuters| 1454326171|David Frost son d...|          betternews|
| false|
|          rotoreuters| 1454326171|Syria: former UK ...|          betternews|
| false|
|          rotoreuters| 1454326171|The forgotten sto...|          betternews|
| false|
|          rotoreuters| 1454326172|British woman con...|          betternews|
| false|
|          kaizoku99| 1454326172|Income sharing ag...|          studyAbroad|
[removed]| false|
|          rotoreuters| 1454326173|Russian Foreign M...|          betternews|
| false|
|          Tectonic_| 1454326173|The most depressi...|          GlobalOffensive|
| false|
|          rotoreuters| 1454326173|Russian lawmaker ...|          betternews|
| false|
|          rotoreuters| 1454326174|Australia's only ...|          betternews|
| false|
|          rotoreuters| 1454326174|Thor 3: Ragnarok ...|          betternews|
| false|
|          rotoreuters| 1454326174|Ofcom warns O2 ta...|          betternews|
| false|
|          rotoreuters| 1454326174|Indian police hun...|          betternews|
| false|
|          jumanji112| 1454326174|The need for psyc...|          Social_Psychology|
[removed]| false|
+-----+-----+-----+-----+-----+
-----+-----+
only showing top 20 rows

```

Operations

```
In [37]: df.select(df.created_utc * 2).show()
```



```
+-----+
|(created_utc * 2)|
+-----+
|      2908652328|
|      2908652330|
|      2908652330|
|      2908652330|
|      2908652332|
|      2908652332|
|      2908652332|
|      2908652334|
|      2908652338|
|      2908652340|
|      2908652342|
|      2908652342|
|      2908652342|
|      2908652344|
|      2908652344|
|      2908652346|
|      2908652346|
|      2908652346|
|      2908652348|
|      2908652348|
+-----+
only showing top 20 rows
```

```
In [38]: from pyspark.sql.functions import log
df.select(log(df.created_utc * 2)).show()
```

```
+-----+
|ln((created_utc * 2))|
+-----+
|      21.790955693332|
|      21.7909556940196|
|      21.7909556940196|
|      21.7909556940196|
|      21.790955694707204|
|      21.790955694707204|
|      21.790955694707204|
|      21.790955695394807|
|      21.790955696770016|
|      21.79095569745762|
|      21.790955698145222|
|      21.790955698145222|
|      21.790955698145222|
|      21.790955698832825|
|      21.790955698832825|
|      21.79095569952043|
|      21.79095569952043|
|      21.79095569952043|
|      21.790955700208034|
|      21.790955700208034|
+-----+
only showing top 20 rows
```

Aggregations

```
In [39]: df.where("not over_18").groupby(["author", df.subreddit]).count().toPandas()
```

Out[39]:

	author	subreddit	count
0	truthfulcoffee	AndroidQuestions	1
1	SuperSaguaro	10cloverfieldlane	1
2	[deleted]	CFB	14
3	rafamorgan	airconsole	1
4	jsntz007	fastloanunemployedau	1
...	...	...	...
127745	Fugeezy	runescape	1
127746	LordKarnox	Warhammer40k	1
127747	[deleted]	swapit	1
127748	Botclone	ImaginaryProRevenge	1
127749	Dovahslayer	DragonsDogma	1

127750 rows × 3 columns

Custom functions

```
In [40]: from pyspark.sql.functions import length

df = df.withColumn("length", length(df.selftext)) # This adds a column

df.where("length > 1000").toPandas()
```

Out[40]:

	author	created_utc	title	subreddit	selftext	over_
0	BlairTheWiseViking	1454326164	(24M) with a DB problem that's going on three ...	deadbedroom	This post will be long as to give detail about...	Fal
1	sleepyechoes	1454326170	Presidential Football Poll Results	CFB	I put together the results in probably the lea...	Fal
2	Daedin_	1454326175	Looking for 3-4 rank 14-18 players to progress...	hearthstone	Hey reddit! Glad to meet you all :) \n \n I'm osc...	Fal
3	Holo_of_Yoitsu	1454326188	[Spoilers] Diamond no Ace: Second Season - Epi...	anime	**Episode title:** To This Side \n **Episode d...	Fal
4	SmellyFbuttface	1454326190	Why doesn't challenge mode change?	DestinyTheGame	I like that they integrated challenge mode. G...	Fal
...	...	...	...	...	...	...
9809	-Boman-	1454326072	Warwick The Fast Ticket Out Of Bronze and Silv...	summonerschool	Hello my name is Boman and today i would like ...	Fal
9810	GoldenJermbag	1454326080	I've finally had my first true high sodium mom...	DestinyTheGame	Before I go tell my woeful tale I will say I l...	Fal
9811	BhiQ	1454326085	FtMs - your thoughts on the male gender role/s...	asktransgender	Hey guys, I'll try to be as respectful as I ca...	Fal

	author	created_utc	title	subreddit	selftext	over_
9812	SpaceyMoogle	1454326087	Freaking out a little about possible risk of p...	TwoXChromosomes	Not really sure if this is the right place. \n...	Fal
9813	PamelaBreivik	1454326125	[ALL SPOILERS] If there is any character that ...	gameofthrones	I understand that I'm probably posting too muc...	Fal

9814 rows × 7 columns

```
In [41]: from pyspark.sql.functions import udf

def splitWords(e):
    return e.split(" ")

splitWords = udf(splitWords)
df.select(splitWords(df.selftext)).show()
```

```
+-----+
|splitWords(selftext)|
+-----+
|[This, post, will...|
|                    []|
|                    []|
|                [[deleted]]|
|[So, waaaaaaaay, ...|
|                    []|
|                [[deleted]]|
|                    []|
|                    []|
|[I, put, together...|
|                    []|
|                    []|
|                    []|
|                    []|
|                [[removed]]|
|                    []|
|                    []|
|                    []|
|                    []|
|                    []|
+-----+
```

only showing top 20 rows

Any dataframe can be used as an SQL table with a syntax similar to that used in other database courses. In fact, this is the basis of the Apache Hive project.

```
In [42]: df.createOrReplaceTempView("reddit")
```

```
In [43]: spark.sql("select * from reddit limit 10").show()
```

```

+-----+-----+-----+-----+-----+
|          author|created_utc|          title|          subreddit|
selftext|over_18|length|
+-----+-----+-----+-----+-----+
|BlairTheWiseViking| 1454326164|(24M) with a DB p...|          deadbedroom|This po
st will be...| false| 3872|
|          DAVIDSPZGZ| 1454326165|EAL: Rebajas Sony...|          ElAndroideLibre|
| false| 0|
|          chasdave1981| 1454326165| Ms Juicy (retired)|          bigasses|
| true| 0|
|          [deleted]| 1454326165|[H] ST MW Aquamar...|GlobalOffensiveTrade|
[deleted]| true| 9|
|          SableArgyle| 1454326166|Something might h...|          OnePiece|So waaa
aaaaay bac...| true| 650|
|          nilimajaveri| 1454326166|Buy queen bed mat...| InternetIsBeautiful|
| false| 0|
|          [deleted]| 1454326166|          Grand entrance|          gifs|
[deleted]| false| 9|
|          a2zlifestyles| 1454326167|9 Cool Indian Hai...|          HairStyle|
| false| 0|
|          dingdong89| 1454326169|Robert Leslie - I...|          BinauralMusic|
| false| 0|
|          sleepyechoes| 1454326170|Presidential Foot...|          CFB|I put t
ogether th...| false| 7831|
+-----+-----+-----+-----+-----+
+-----+-----+

```

Assignment question

Get the users who have uploaded to Reddit more than 10 posts of over 100 characters.

Answer:

```

In [68]: spark.sql("""
SELECT author, COUNT(*) as post_count
FROM reddit
WHERE length > 100
      AND author != '[deleted]'
GROUP BY author
HAVING post_count > 10
""").show()

```

author	post_count
boardgamerecommender	12
dota2matchbot	230
soneo143	14
JmodTracker	11
lolretkj	12
gyfaglover4	242
AutoModerator	616
Jesupekka	21
autotldr	221
Chabombs	13
cruyff8	83
removalbot	305
unityrufa	11
ShitSeattleSays	11
leaopeng	102
ccmanga	57
MartyMcfly6	15
james4398	15
asamale	12
victorantos2	12

only showing top 20 rows

SQL operations

How to declare a view from a Dataframe

```
In [44]: df.createOrReplaceTempView("reddit")
```

```
In [45]: spark.sql("select * from reddit limit 10").show()
```

```

+-----+-----+-----+-----+-----+
|          author|created_utc|          title|          subreddit|
selftext|over_18|length|
+-----+-----+-----+-----+-----+
|BlairTheWiseViking| 1454326164|(24M) with a DB p...|          deadbedroom|This po
st will be...| false| 3872|
|          DAVIDSPZGZ| 1454326165|EAL: Rebajas Sony...|          ElAndroideLibre|
| false|      0|
|          chasdave1981| 1454326165| Ms Juicy (retired)|          bigasses|
| true|      0|
|          [deleted]| 1454326165|[H] ST MW Aquamar...|GlobalOffensiveTrade|
[deleted]| true|      9|
|          SableArgyle| 1454326166|Something might h...|          OnePiece|So waaa
aaaaay bac...| true| 650|
|          nilimajaveri| 1454326166|Buy queen bed mat...| InternetIsBeautiful|
| false|      0|
|          [deleted]| 1454326166|          Grand entrance|          gifs|
[deleted]| false|      9|
|          a2z1lifestyles| 1454326167|9 Cool Indian Hai...|          HairStyle|
| false|      0|
|          dingdong89| 1454326169|Robert Leslie - I...|          BinauralMusic|
| false|      0|
|          sleepyechoes| 1454326170|Presidential Foot...|          CFB|I put t
ogether th...| false| 7831|
+-----+-----+-----+-----+-----+
+-----+-----+

```

Assignment question

Obtain the users who have posted in reddit more than 1k characters in any subreddit with SQL (without using any column named length)

Answer:

```

In [70]: spark.sql("""
          SELECT author, COUNT(*) as post_count
          FROM reddit
          WHERE length(selftext) > 1000
          AND author != '[deleted]'
          GROUP BY author
          HAVING post_count > 10
          """).show()

```

	author	post_count
	boardgamerecommender	11
	soneo143	14
	AutoModerator	89
	autotldr	221
	cruyff8	64
	removalbot	38
	unityrufa	11
	leaopeng	102
	james4398	11
	scamcop	15
	juriah121	102
	skdgldgajdgkj	11
	FFBot	12

Other libraries

Beyond dataframes, we can find other libraries that also rely on Spark...

In [46]: `import pyspark.pandas as ps`

```
# Disable Spark ANSI mode for pandas-on-Spark compatibility
spark.conf.set("spark.sql.ansi.enabled", "false")

# Convertir a pandas-on-Spark
kdf = ps.DataFrame(df)

print(kdf.head())
```

```
/usr/local/lib/python3.12/dist-packages/pyspark/pandas/__init__.py:43: UserWarning: 'PYARROW_IGNORE_TIMEZONE' environment variable was not set. It is required to set this environment variable to '1' in both driver and executor sides if you use pyarrow>=2.0.0. pandas-on-Spark will set it for you but it does not work if there is a Spark context already launched.
  warnings.warn(
```



```

author created_utc
title subreddit
selftext over_18 length
0 BlairTheWiseViking 1454326164 (24M) with a DB problem
that's going on three years. deadbedroom This post will be long as to
give detail about the situation. I'm not looking for pity but some more answers a
nd to possibly help others out there who find themselves in the same situation.\n
\nAs stated. I am a 24M. My other half is a 22F. We'll reach six year together on
February 19th. The DB has been going on since 2014. We have sex once a year. Mayb
e twice. That's a stern maybe. I can confidently say I love her and would rather
not be with or have anyone else in every sense of the meaning. I've been with her
for six years. Obviously there's love there.\n\nWhen we first started dating in 2
010 the sex was very frequent and very hot. Multiple times a week and just fun an
d wild. There was "romance" there in terms of there also being slow, passionate s
ex. We had a blast. Eventually overtime things started to taper off. Rather quick
ly too. It got to the point where we have sex once a year. Let me make it clear t
hat sex isn't everything to me. I love this lady. She knows things about me in wo
uldn't dare allow someone else to ever know. She's stuck with me. However, sex is
a mist in every relationship. For emotional reasons. It bonds couples. It strengt
hens each other. It relieves stress and it let's the other person know you flat o
ut still find them sexy as shit. Hell, when we're 80 I'd still want to have her.
Her tits could drag on the floor and I'd still hit it. I love this chick. But w
e're not really what we used to be in the sex department. It's gotten bland and b
oring majorly. After a while and shortly before the drop off, I always had to sta
rt it. Always had to be on top. She didn't want to do the work. She got real lazy
about it. I've tried spicing things up. Offered to try new positions, try toys, d
ifferent places, anything that could set fire to what we used to have and go back
to having those fun, naked moments. It doesn't work. She gets embarrassed. We use
d to do these things but not anymore. Hell, she doesn't like giving blowjobs anym
ore. She thinks they're gross but she used to give good ones.\n\nI'm not sure wha
t it is. I've tried talking to her about it. Most of the time she shuts down and
doesn't wanna talk or walks away. I got her to tall a little once and her reasoni
ngs were there was no time due to work and that she just isn't a sexual person. B
ullshit. On her time off she spends hours watching TV shows and reading and she i
s sexual because she used to like doing some crazy shit. I don't know what to do.
I've tried initiating and it never works. Open relationships are not an option be
cause I don't want anyone else and I really don't want another guy putting his sh
itty little hands on my lady. I know she'd feel the same way. We've never been on
es for that kind of shit and we'd never cheat on each other. That's been made ver
y clear before.\n\nIt's difficultly being the guy in the situation. You don't wan
t to come off as a pig or as selfish. You want to be as respectful as possible ab
out it but it always sorta comes off that way. I know I'm not wrong here thinking
once a year is batshit crazy for a relationship. Especially us being so young. I
t's also difficult because I'm not usually once to be insecure about shit. I'm a
6ft 230lbs guy with a monster beard, covered in tattoos, and drinks beer. Being i
nsecure isn't my thing but this has got my head spinning. I feel that maybe she n
o longer sees an attraction in me anymore or even worse, what if there is another
guy that I don't know about. It's insane how much it has me freaking out because
I am in love with this little lady and I want to marry her, grow old with her, do
all of the romantic and stupid things couples do. I don't want it to fall apart b
ecause of sex. That's just not an option.\n\nWhat do you all think? Am I nuts?\n
\nTLDR: Six years together and have sex once a year. Love the lady to death and w
ant to get things back on the naked track. False 3872
1 DAVIDSPZGZ 1454326165 EAL: Rebajas Sony, hasta un 50%
en dispositivos y accesorios ElAndroideLibre
False 0
2 chasdave1981 1454326165
Ms Juicy (retired) bigasses
True 0
3 [deleted] 1454326165 [H] ST MW Aq

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uamarine Revenge [W] 38 keys GlobalOffensiveTrade
[deleted]      True      9
4      SableArgyle  1454326166  Something might have been missed a while back
(Manga Spoilers Chapt 814+)      OnePiece
So waaaaaaaay back in chapter 581 we got a glimpse of something that was really u
nimportant at the time. Bonney's Ship. It has a cake theme. As seen here. http://
i.imgur.com/IGraoYM.png \nSo with this possible new information (not sure has any
one else brought this up before?) this could mean we at least know one of the pos
sible guests for the wedding. \n\nOf course this could also imply that Bonney is
one of Charlotte's many children which would be an interesting revelation.\n\nIf
you're willing to make a bit of a reach as well, Bonney has pink hair, so does Lo
la, so maybe that theory holds a little more water? \n\nWhat do you guys think ab
out this?      True      650

```

```

In [47]: kdf["sumChars"] = kdf.selftext.str.len()
res = kdf.groupby(["author", "subreddit"]).sum()
res[res.sumChars > 1000]

```

Out[47]:

		created_utc	t
author	subreddit		
-Boman-	summonerschool	1454326072	Warwick The Fast Ticket Of Bronze and S
-Dee-Dee-	quilting	1454345640	looking for nfl fabric words for book qu
-GD-	NoFap	1454343498	Adding NoFap to improvement
-KbR	GlobalOffensiveTrade	2908648008	[STORE] All kind of s higher and lower t
-Leafious-	boogie2988	1454340974	Boogies comments ab the Fine Bros on stre
-Sasang-	nosleep	1454347189	Deep in the Adirond Mounta
-Wubbalubbadubdub	Cypher	4362928699	Balance (Criti Requested)Unti (Critiqu
-_-l-l-_-	Minecraft	1454309418	Help with squid farm
-little-one-	sex	1454302437	[Question] Sex with boyfriend [22M] is pe
-ultraviolence-	RandomKindness	1454317562	[OFFER] [US] home de
010116	MechanicAdvice	1454338987	Having problems with car dealer's mechan
OILERS	pathofexile	1454354067	Self-cast hybrid Blade Some gear/stat ti
Orontes	Netrunner	1454354783	A "my first tournament" p
0x0af	cscareerquestions	1454339320	BS Information Security MS CS worth
127crazie	clevelandcavs	1454320013	How did the jerseys cha between 2011-2014
13lack12ose	relationship_advice	1454300914	[18M] I'm not happy in current relationsh
17496634303659	techsupport	1454339880	Girlfriend got water onto laptop... It w
1975inanutshell	relationships	1454311585	My [24/M] boyfriend br up with me [22/F]
19832012	occult	1454299923	Past Life Is
1ManGnarmy	Walther	1454355336	PPS m2 - 400 round
1Q98	Surface	1454304490	Help in choosing betw the M3 and I5 4gb 1
222Fusion	thedivision	1454305637	Some general questic hope someone has ans

		created_utc	t
author	subreddit		
22Anonymous22	creepypasta	1454297936	Help me understand Japanese horror cryp
22deviousbroadwaybou	offmychest	1454305255	He deserves everything has got
23458273465982736	relationships	1454284872	Me 33M with my 25F gf c years, constant ne
23andcounting	Herpes	1454287836	Kind of lost right r
2806421200	tipofmytongue	1454298738	[TOMT] [BOOK][
29402	USMC	1454304157	Trying to help a friend t honorably serve
29425	StackAdvice	1454304480	Spaced out feeling on da don't take noope
2Countries1Problem	relationships	1454326825	UPDATE: Flying to visit [26F] boyfriend [
2dP_rdg	malefashionadvice	1454299087	Feedback request for fitment of these d
2good4hisowngood	thedivision	1454352618	Some continuity issu noti
2much2know	MakingaMurderer	1454296391	Calls to dispatch stevenaverycase.com. Wh
2ofSorts	insurgency	1454307352	What should I expect fr this Gar
3veryTh1ngWr0ng	SuggestALaptop	1454340787	2D animat
412bird	GlobalOffensiveTrade	4362927080	[H] Keys [W] Nice f playskins[Store] som
420Throw69away420	tifu	1454299195	TIFU by getting drunk for first t
42kinda-human	LifeAfterNarcissism	1454352313	Sweeping the deck -- ACoN mom
44davi	Glitch_in_the_Matrix	1454294768	Don't know if this belo r
4benny2lava0	malehairadvice	1454289240	One of my more hor friends says its time
4everaloam	relationships	1454355430	Me [29/F] with longt (10+ yrs) friend [27/
5ft0lady	relationships	1454294808	Me [26F] with my boyfri [28M] 3mo/duratio
612Shane	bodyweightfitness	1454301611	Opinions on weekly schec set up: Looking

		created_utc	t
author	subreddit		
612pab	BMWi3	1454334805	OTA updates BMW I3. Is seric
725Cali	raisedbynarcissists	1454348353	Nmom retold someth from my college days a
76SUP	tfour	1454291344	Fighter Submissio
7H3_d0c70r	electronic_cigarette	1454329766	Temperature Control vap and you! (some har
7ripp	RecruitCS	1454353188	[NA] Central/East Co SMFC/Global Team Loc
80sMR2	polyamory	1454307197	Struggling: Where do I now in this relatic
9000Proof	DentalHygiene	1454311800	Bleeding gum and my l tee
99hotdogs	boston	1454352766	10 Observations from a r Bostor
AAL314	asoiat	1454302837	(Spoilers Main) Did Rhae ever own a Valyr
ABrainInAVat101	CastleClash	1454326355	Aries - Revite 5 Still Cruc
ACIDLIF3	depression	1454291732	It's getting wo
ADVentive	hometheater	1454341375	Mount the TV ov window, or sit it on a ca
ADaddyHansen	techsupport	1454299623	How do I create a windo 10 recovery drive
AMuseXD	PurplePillDebate	1454287017	Common RP Argume One BP's Respor
ASkylineDiver	crochet	1454339063	Hookers of Great Bri (England, Scotland,
AVG_AMERICAN_MALE	awardtravel	1454295679	Can't find 12.5k AA miles my trip in Aug
AWesome_Sawse	personalfinance	1454286734	First time independ contractor, late to
A_1000	techsupport	1454307134	Having trouble format new ?
A_Gris	Warframe	1454306366	It'll never happen to right? 3 years in :
A_Pushover_Dad	dirtypenpals	1454340976	M4F - Cheating, M Games, Depravity... Eve
AaronWithAQ	UHCMatches	2908580208	Feb 01 02:15 UTC N WizzedPlays's #16 - FF

		created_utc	t
author	subreddit		
AbaddonSF	TryingForABaby	1454304672	Getting Pregnant and sl aids h
AbbyKatDPP	dirtypenpals	1454341895	F4F Fut
Abe_Vigoda	SuicideWatch	1454324524	I'm going to a fun tomorr
AbioFlesh	gamemaker	1454345265	How to create uneven ter and adapt moveme
Abno_csgo	GlobalOffensiveTrade	4362959652	[H] Karambit Doppler Karambit Blue Steel
AboutNerf	Nerf	2908651949	Auto-Tracking Jolt Turret only \$40Offic
AbsoluteFake	depression	1454289199	(18) wouldnt mind dy
AbstainLoL	summonerschool	1454322990	Need tips for analyzing f
AceShadowX	GlobalOffensiveTrade	5817256309	[Store] IBP Katowice ite Glock Fade IBP, I
Achierius	HistoricalWorldPowers	4362972001	The Icelandic F SailsFormal Annexation c
Acquiesce67	hackintosh	1454339067	gpuRestart // Game free then continues af
ActSci94	relationships	1454308979	Should I [M21] continue relationship with
AddontheDespoiler	Advice	1454307125	Need help on dealing wit bad break up. He
AdfertMortem	CasualPokemonTrades	1454324212	FT a whole lot o' shinies: Shiny Skarmory/!
AdmiralKird	asoiat	1454340505	(Spoilers Everything) Time to Recognize
AdrianWillis22	watchinganime	1454338068	Nagi no Asukara Episoc
Aerest	pathofexile	1454326884	Can we make a list of CV mo
Aeris_24	Warframe	1454325363	I'm quitting this game here is why. No, y
AgentKittyfeets	childfree	1454348747	Oh, sure. You comment N
AgentLove	bloodborne	1454318016	*Lore Spoilers* Brain F
Agent_Jesus	math	1454337754	Does there exist a num (or a set of numbe
Agides	SandersForPresident	1454347527	Bernie puts Verizon, Ban America and Boe

		created_utc	t
author	subreddit		
Agilaz	killerinstinct	1454336988	Playing 3-year catch
AgravatedArdvark	300zx	1454340238	To TT or not to
Agresan	Starcitizen_trades	1454325259	[WTS] LTI Ships & Upgrades & ADD-C
Ahnaful1994	Ahnaf	24723170195	48000000XP in Magicl for a dragon helmLevi
Ahnedonia	raisedbynarcissists	1454291018	My Naunt made my 13y me cry and took pictu
Aidensman	whowouldwin	2908629607	[Bloodmatch] The Aveng (MCU) VS Accelerat
Aigo7yie	Parabola	1454322552	What about playing proprietary games wit
	freesoftware	1454321194	What about playing proprietary games wit
Ainbow_Dish	tf2shitposterclub	2908694980	why did mr.paladin cross roadWe should m
Airum	photoshop	1454293316	Help! Grid lines appear in shadows after
AishiteruAika	GlobalOffensiveTrade	4362995166	[Store] Blue P3/,P2,Fade,1 Best F.v Serpe
AkaashMaharaj	pbsspacetime	1454288729	Perplexed about the t paradox in a curved
Aktienkurse	Verschwoerungen	1454355450	Gladio / Stay Behind / Gla
Alansmithiee	amiiboSwap	1454312155	[US] [Glendale and Westla CA] [H] (001),
Albigularis	MotoUK	1454344801	Delive Drivers/Motorc Delive
Aldael	razer	1454332035	Razer Swtor mo doubleclik
Alec_Ross	IAmA	1454348740	I am author Alec R former Senior Adviso
AlexG9090	GlobalOffensiveTrade	4363027617	[H] AK Aquamarine/Vul MW [W] Keys[H] AK Ac
AlexTheTownPump	SquaredCircle	1454351937	WWE Axxess worth gc
Alexanderdaawesome	learnprogramming	1454312385	I discovered arduino week. today I playe

		created_utc	t
author	subreddit		
Alicia97xx	Roleplaykik	1454322625	F4F looking for some sex
All-StarJohnScott	NHLHUT	4363036988	[XBOX1] would anyone interested in a givea
All1nAlex	MakingaMurderer	1454349541	Two questions - hoping clarificat
AllHawkeyesGoToHell	CFB	1454294049	FBS FCS-Style Play
AllYourBass6174	AndroidQuestions	1454325686	Google-related apps for cl
All_Luck_No_Skill	leagueoflegends	1454291927	Is it just me or does the r system just ma
AllenThePrettyCool	halo	1454327891	Thoughts on Halo 5, What I Feel Should H
AlliedKhajiit	Guildwars2	1454291593	The skills that can and c be used underw
AllisGreat	GlobalOffensiveTrade	1454296397	[H] Sexy 4x Kato' 14 HC playskins [W] Key
AlphaGangplank	summonerschool	1454340193	Why is Gnar so bad at mome
Alsnake55	DvZ	1454285542	Terrain For A Map- Hanc Off The Tc
AltMom	ankeny	1454295598	Weekend Update: Plot T Bookst
AlwaysBeNice	awakened	1454285506	A part of my favorite b on 'being' instea
AlwaysSomewhereElse	Frugal	1454288021	I may have just finished last box of ice
AmaruGalvarino	Anarchy101	1454341073	How do i explain trotskist gf that the s
AmberDingMe	DarkNetMarkets	1454295278	Product Review - Whiteout (wee Ganeshl
Amberftw	SmiteLFM	1454356409	[PC] [ALL] Konvict Gami Teamspeak 16+
Amuseph	travel	1454356757	Excellence Punta Cana Sandals Jamaica For
AnExquisiteCorpse	stopdrinking	1454350290	Sobernaut reporting
Anarchanoid	wow	1454356541	Rate That Instance - Day
Andre_SSS	GlobalOffensiveTrade	2908636799	[H] Some Ak skins with k 14 stickers Less

		created_utc	t
author	subreddit		
AndrewRichmo	nonfictionbookclub	1454342185	Manufacturing Conse Chapter 7 & Fini
Angellions	exjw	1454303441	Seriously? We need thi s
Animal31	gamedev	1454304528	Mastermind Algorithm v a single in
AnimeMusicLove	Advice	1454297535	College Stud
Anon_Syndicate	buildapcforme	1454331355	Gaming/Media consumpt
AnonymousTurker	electronic_cigarette	1454292588	AnonymousTurker's Free Very Cheap Juice
AnotherAngryAussie	couchsurfing	1454345887	Account Locked Again (t ime)... time to mo
AnotherFewMore	SCCM	1454291255	Office installat peculiariti
Ant-n	Bitcoin	1454357513	Soft fork Segwit and SP attack weakn
Antetokounmpo	MkeBucks	1454287851	Update 1/31/2
AntonioOfVenice	atheism	2908705456	NECSS' Steven Novella hypocrite (one o
AphelionFive	GlobalOffensive	1454327487	Casual Player Looking Become Competitive &
AppenWAH	workathomejobs	1454320129	Urgent Opportunity: Fre Canadian Native Sp
ApplejacksAndBoners	leagueoflegends	1454313044	I'm finding that mut people equals greater
Applicant123	GetEmployed	1454351751	Can I list volunt experience as work expe
ArabJedi	tifu	1454349323	TIFU by saying a word Russ
Argentis	jewelry	1454311912	Just received a HL amount of face gemsto
ArkSurvivalEast	playarkservers	1454299998	-->FRESH WIPE Filling FAST!<-- (Tod
Arrow_Raider	sysadmin	1454331669	Windows 10 Lock Scr
Asatania_GM	DnD	1454345105	Time is almost up on fully funded Kickst
Astagirl	Anarchism	1454306203	Announcing A New Pro
AstonishingSpiderMan	NatsuDragneel	1454304181	Natsu Dragneel Comm Fa

		created_utc	t
author	subreddit		
AtomV3	Pins4Skins	1454330160	[H] GOOD PRICES & REP - 3 pins at 16 key
AtomicCoyote	TalesFromRetail	1454289454	Any woman should kr
AtomicSamuraiCyborg	Pathfinder_RPG	1454354789	[Skull & Shackles] they're going to tl
AttDominate	learnpython	1454349571	Im doing "UTAx: CSE13 Introduction to Pro
AuahDark	SchoolIdolFestival	2908640263	[Other] Help extracting r story background
AudaciTEA	mcservers	1454295628	AeriaCraft [SMP] [Van [PVP] {Whitelist
AutisticJaffaCake	depression	1454285226	My happines decompos
AutoModerator	ACTrade	1454310168	[PSA] Daily PSA Thread February 01, 2
	AirForce	1454328396	Weekly Newbie Threa Post questions about
	Anxiety	1454316376	Welcoming Newcom & Free Talk Thread -
	Aquariums	4362935317	[Auto-Post] Weekly Quest Thread! Ask /r/ac
	Austin	1454306833	Stuff To Do In Austin - W of 02
	CasualPokemonTrades	1454321080	Good Manners and Comn Court
	CaveOfOrigin	1454324609	Daily Stuff Thr
	DFO	1454338994	Weekly /r/DFO LFG Threa February 01, 2
	DIY_eJuice	2908656617	Weekly New Mi Questions Thread - Wee
	Denver	1454355152	Things to Do in Denv Week of February 0
	DestinyTheGame	1454331845	Daily Thread - Mer Mon
	Diablo3witchdoctors	1454339010	Consolidated Reroll Gear Advice Thread f
	Dogtraining	1454292135	02/01/16 [Loose Le Walking Virtual Worksh

		created_utc	t
author	subreddit		
	FashionReps	1454339104	General Questions Monday February 01, 2016
	Homebrewing	2908656974	Daily Q & A! - February 01, 2016Sitrep Monday
	HunterXHunter	1454356856	Hunter × Hunter 19 Episodes 63 & 64 —
	IndieGaming	1454320999	Self Promotion Megapost 01 February 2016
	LGBTeens	1454306759	[Me Monday] - February 2016
	MMORPG	1454328255	Weekly LF MMO Suggestion Thread- February 01, 2016
	MechanicalKeyboards	1454320951	/r/MechanicalKeyboards What Keyboard, Switch, etc?
	Megaten	1454288573	Game of the Month February - Shin Megami Tensei
	NHLHUT	10180236995	Mentor Monday - Weekly Question Thread - February
	OpTicGaming	1454342770	[MISC] Daily Discussion Thread (February 01, 2016)
	RateMyAudio	1454332129	RateMyAudio February, 2016 - Monthly audio suggestions
	RedditKnightsRCS	1454349749	Monday War Callout Thread
	Romania	1454285023	Întrebați /r/Romania Ori 01.02.2016
	RunnerHub	2908656142	The AAR Megathread 01.02.16 to 15.02.16The
	VinewoodLocos	1454326354	Crew Stats - January 2016
	WeddingPhotography	1454331899	Wedding Photo Critique (Official Thread): PC
	aldub	2908606949	[SPOILERS] AIDub I Discussion Thread: February
	audiobooks	1454324690	Welcome to /r/Audiobooks Please read before
	audioengineering	1454331807	Recommendation(w should I buy?) Mondays -
	audiophile	1454331374	Daily Purchase Advice Thread (2016-02-01)

		created_utc	t
author	subreddit		
	buildapc	1454328100	Simple Questions - Febru 01, 2
	churning	39266579573	Official Alaska Airlines \ SignatureOffic
	dfsports	2908606056	Daily CBB Discuss (February 01, 2016)Dai
	dvdcollection	2908606228	Digital Code Exchang (February 2016)For S
	electronicmusic	2908667221	[OFFICIAL] Recommende You Like Thread -
	fandomnatural	1454347952	Fandomnatural's Mon Introduction Thre
	hiphopheads	1454340732	Moronic Mondays - Wee Question Thread - Fe
	investing	2908656290	It's moronic Monday, y chance to ask any
	linguistics	1454331630	This week's Q&A thr -- please read be
	loseit	2908605698	24-Hour Pledge - February 2016Motivat N
	madisonwi	1454328163	MONTHLY JOBS POST - V needs work? Who's gc
	myNBA2KMobile	4362930888	Auction House Threa February 01, 2016Wee
	nba	1454303039	Daily Sidebar Im Selection Thr (2016.02
	onlinemovieplaylists	30540757409	It's Fun and Fancy F Discussion! (Februar
	powersofmiddleearth	4362855555	Sunday is over! RP pe now allowedWeekly E
	proED	2908670758	Weekly Stats Upd February 01, 2016Daily I
	punk	2908688673	Discussion Thr 01/02/16New Music Mon 0
	shadownet	4363045751	Official Jackpoint Thr <01/02 - 07/02&
	stunfisk	1454306791	Q&A Thread: Are new? Got questions?

		created_utc	t
author	subreddit		
	summonerschool	1454299580	Weekly Replay Rev Thread: Week
	ttcafterloss	4362931017	/ttcafterloss TTC D Discussion Thread -
	visualnovels	1454349763	What are you reading? Untranslated edition -
	yugioh	1454328014	/r/YuGiOh Basic and New Q&A Thread -
AverageJoeAudiophile	BudgetAudiophile	1454287035	Monthly for sale p Febr
AvoidingTheENT	AskDocs	1454288179	Explosive Sinu
Aymanbb	GlobalOffensive	1454323853	Graphics card temps hea affect my mo
Ayyra	dota2circlejerk	1454308825	Arteezy and grandgra thoughts about team
Azathoth321	DiamondHunt	2908706250	The building cost of MBV too high?Dragon /
Azebu	archeage	1454346730	Few questions about hea and PvE cont
Azhf	BasketballGM	1454307995	Hoop City Season 1 BasketballGM Narra
AzraelDirge	TalesFromTheMilitary	1454353769	The Legend of A1C Dips Pt 4: A1C Dipshit
AzteCGG	RecruitCS	2908683921	[EU] UK, Looking for a te to play leagues
B0BtheDestroyer	asoi	1454345443	(Spoilers Extend Speculation on the Three
B3qui	LongDistance	1454286146	Entangled with a boy fr Ecuador, a web of b
BEARDown4Midterms	DetroitRedWings	1454351742	Rest of the season outl
BH_Andrew	RainbowSixSiege	1454342695	A series of unfortun events (funny story a
BZimmiZB	thedivision	2908684981	This game is in need o crouch mechanicWha
BaBopByeYa	TalesFromYourServer	1454357647	""Tis the season!" - My st of that one tin
Baba-Boom	Clojurescript	1454350725	How to do Token Auth (
Bacio	theyear	2908713376	End of week 4Week 4 d and Week 5, da

		created_utc	t
author	subreddit		
BackHandAces	GlobalOffensiveTrade	1454356709	[Store] Some High Skins/Knives! Also Bu
Bacon2014	denverjobs	1454352845	[HIRING] Mid Leve Developer 75K Denver,
BaconWrappedKitten	rape	1454310650	A life time of abuse effecting my day t
BadassRipley	GFD	1454325283	[Game] Interested in join a FTB Minecraft
Badnapp420	buildapc	1454355090	[Build Help] - BudgetPC GFs Dad - Specif
BadonkadonksGaming	elderscrollsonline	1454315240	[Discussion] More Ambie sou
BagOnuts	PoliticalDiscussion	1454329927	[Megathread] 2016 Ic Caucu
Bainy_	GlobalOffensiveTrade	4363007193	[H] 0.009 FV m9 tt [W] keys[H] 0.009 fv n
BakeMeA	techsupport	1454308759	Asus Anti-Surge protect triggers every tin
Balkazzaar	Alienware	1454348556	Alienware 15 r2 GPU/Scr questi
Ballerman4452	Atlanta	1454348816	Looking for a room in
Ballistic87	NoFap	1454344849	Uk nofapers, anyone fee like shit during
Baloth	starcitizen	1454337483	Ballistics penetration cc use a re-w
BamaboyinUT	tahoe	1454307165	Coming this week. Hav few questic
BananaHamSalad	flashlight	1454293500	Replacing a 3.7v lithium with a permane
BanjoBilly	randpaul	1454301216	So how exactly can R surprise everyo
BarbaricZar	GlobalOffensiveTrade	4363043484	[H] *Collectors Item* ST 5-7 CH #2 Patter
Bardock1233	gameswap	1454296574	[USA] [H] Various Titles (3 DS, PS1, N64,
BargnaniDefense	FifaCareers	1454345856	Journeyman Career vs Yo Acade
BaritoneAssoluto	ClassicalSinger	2908695012	[Rep] Recommended vc literature/repertoir

		created_utc	t
author	subreddit		
BaronDeSpireal	offmychest	1454324109	No, the Fine Brothers did turn into cash
Bassbomber	UWMadison	1454350899	Accept
BatiH	The_Donald	1454316165	Four years after Trump ' elected: A new Ro
BayesianJudo	csshelp	1454288664	Any creative way to m parent select
Bbilbo1	tabletopsimulator	1454337537	Open Call to Online Art Critics, and St
BearXW	HalfLife	1454308325	Black Mesa is mind-blow
Beardalelist	servers	1454304892	I'm new here. Reckon ' could help me with n
Bearjewpew	NoFap	1454304950	Thank you random inter strang
BeastG01	PostPreview	1454332371	Test
Beatnik_Exploit	keurig	1454289874	What am I doing wrong, Keurig 450 keeps t
BeingMC	UHCMatches	2908590058	Feb 01 02:45 UTC M Being's #5 - cTo3 - Cu
Benevolay	swtor	1454307797	Unsure How To Proc
Benjenzo	anime	1454296121	[Spoilers] Koyomimonoga - Episode 4 [Dis
Bennators	NHLHUT	7271471258	0-0 after 10 mins of all game!?ASG 93 T
Bennguins	NHLHUT	1454293489	Jamie Benn Rev
BennyDPP	dirtypenpals	1454351982	[M4F] The Billionaire's :
Beselod	bloodborne	1454333964	As someone who ha played in months, how
BestLeeNigeria	GlobalOffensiveTrade	4363036461	[H] 2 KNIVES: Bayc Doppler FN, Flip Ultra
Beware_of_Snakes	ShittyPoetry	1454344864	Fuck Your Future and T Up Your Cr
Beythoven	GlobalOffensiveTrade	4362973158	[Store] Lots of quality K '14 Holo guns,
BiSlyGuyy	lgbt	1454329954	Need Advice for behavi changes around frie
Bicyclestories	sex	1454292923	[Advice] I [19M] completely unable to m

		created_utc	t
author	subreddit		
BigBooty-Milf	dirtypenpals	1454301195	F4M - Big Booty Milf fuck behind her husband
BigBroagogo	dirtypenpals	1454294252	30 [M4F] - Loving brother/babysitter looking for
BigBurlyBear	DarkNetMarkets	1454309535	People don't care about "stealth" anymore
BigMik_PL	gaming	1454345213	The Division BETA is a proof of how much we
	thedivision	5817245034	Few things we don't appreciate enough in the
BigThickAss	dirtypenpals	1454300531	F4M - I want 'The Rock' fuck my big
BigTxT	indiegameswap	1454352381	[H] Games (with Cards) Cards or Games or
Bingjer	AskDocs	1454351284	2 month cough + other sympto
Bio_Hazardous	buildapc	1454333935	[Build Help] A pretty good deal on CPU and
BionicPotato	buildapc	1454297303	[Build Complete] Finally completed my build
Bipartner	AskFeminists	1454345149	X-post from R/bisexu
Bittersweetreality	FemdomCommunity	1454312973	My bitch and I are officially and I wouldn't i
BlackLantern69	dcrp	1454349919	Blackest Night Falls From S
Blackshell	dailyprogrammer	1454339746	[2016-02-01] Challenge # [Easy] Sailors ar
BlazedBuddha	vegan	1454303083	Organic, Vegan Soil & Compost Teas! Please
BlazerPlays	cats	1454350764	[Help] My cat is terrified the new c
Blazing_Right_Hand	NoFap	1454318017	Morning Wood iss
Blee10	SquaredCircle	7271577985	The moment that made me Batista fan and th
Blinsin	Flyers	2908682745	Summary Sunday: I For Again EditionThe 4/
BloodyRahu	playarkservers	2908616724	Class M Needs to Expand Gene Pool!!! Clic
BluePianist	greece	1454338020	Κριτήρια για να μπει κόμμα στη Βο

		created_utc	t
author	subreddit		
Bluejay888	stopdrinking	1454320061	CHECK-IN Monday, Febru 1, 2016 - I will n
Bo98	ScottishFootball	1454345492	SPFL + Scottish League C Roundup: 29th - 3
	soccer	1454345686	League roundup: SPFL Scottish League Cup (
BobDoleOfficial	EliteDangerous	1454316378	The Plight of Crashy Sma (Distant Worlds
Bohrdumb	climbing	1454334156	Belay Technique Questi
Bond_Ionic_Bond	Warthunder	1454357738	War Thunder etique
BooHoo_WhinyBitch	weddingplanning	1454348417	Underwhelmed with food at our dream venu
Borne2Run	empirepowers	2908581905	[EVENT] The Ottom March for Tabriz[EVENT]
Bound_Internal	relationships	1454298082	Me (M14) and my girlfri (F14) broke up fo
BowlOfSpaghetti	buildmeapc	1454339025	Gaming PC 1,000—1, with a ca
Brainfreezdnb	buildapc	1454313378	[Build Help] From noteb to Gaming
Brake666	travel	1454356262	Next January trave through asia and US ar
Bralth	reloading	1454304151	Why are reloaded rou "bette
BravoGangUS	Rockband	1454338856	DLC Week of
Braydawg9	relationships	1454356253	Me [21 M] with my lo distance girlfriend [
BrazilianButtPirate	peloton	1454309879	Outlandish Predictio February 2
Bright-N-Salty	Christianity	1454353401	Do All Lives Matte Christian Organizatio
BrinkBreaker	EnorousPact	1454289896	The Upin Syst
Bristolian	Unity3D	1454346048	My colleague made a sim html search field
Bro7695	CFB	1454339930	(Day 9/62): What do fans think of the BY
Brochanlee1992	school	1454328817	Every body is friends wi

		created_utc	t
author	subreddit		
Brosephski	DotA2	1454313736	How do you motivate y team without shamin
BrotriousNIG	ynab	1454337845	[nYNAB] What happ when one spends from a
BryDavok	electronic_cigarette	1454342679	Torn between Topbox M and Cuboid with Gr
Buck3tbot	GlobalOffensiveTrade	1454357480	[Store] #6 Exchange ST K P2, FN Flip Dopp
BuddhistBreadLoaf	4USXUS	2908667519	H.R.3632 - Stop Arctic Oc Drilling Act of
Bukkake_Tsunami	ynab	1454319234	First month of YN
Bulbastone	OnePiece	1454352321	Luffy to kill Shanks? The
Bulkforlife66	dirtytenpals	1454341648	[M4F] - looking for a s minded older woma
Burstjoe	MonsterHunter	1454332720	TIL an easier way to Gu P
BusinessPickles	LegalAdviceUK	1454337884	Business advice on foldir company and res
ButchMFJones	montgomery	1454311893	Two weeks a Montgomery music Andrew Le
ButterflyRS	ButterflyRS	29086544063	Levelled Invention.Levelled all s ov
Buzzismydog	trees	1454292756	NEED to share my story v this commu
BwoiToy	FIFA	1454321452	Lag is the biggest prob in this game whic
ByAllMeanss	AskHR	1454351166	I feel like I'm getting trouble for issue
Bytonia	Leathercraft	1454339365	straight vs rounded skiv bl
C0mpass	AyyMD	1454297647	how 2 revive the dank
C0mplx	acrl	2908669886	Season 5 Signups are H (Updated Rules, D
CBNathanael	wicked_edge	1454341456	Best shave in a while, had to break most
CClossus	LyricalWriting	1454286409	Hey, I'm new to lyr writing, and I want y

		created_utc	t
author	subreddit		
CEverii	personalfinance	1454344513	I have a weird 1C situation, and my local
CMDR_ShodFir9748513	ElitePatreus	1454319015	To the Imperial CMDRs v XBone tech in Keni
COUTS_132	summonerschool	1454290808	Laning vs Malp
CR4allthethings	running	1454287986	Looking for support for family of a fallk
CRRZ	golf	1454341141	Took a lesson and working on my short gam
CSGO-Penny	Pins4Skins	1454318777	[H]Pins [W]Keys/Items- REAL MONEY TRAE
CTHULHU_SPEAKS	dirtypenpals	1454331392	42M4F - Corrupting Innoc
CVeeks	malehairadvice	1454338587	Second try with gett advice, let me be mo
CYCOABHI	buildapc	1454296261	[Build Help] Partpic saying somethings wr
Cabooseaholic	hardwareswap	1454312866	[USA-CA] [H] Complete build with monitor
Calavente	noveltranslations	5817326527	[JP] A Certain Middle-A Man's VRMMO Acti
Callahad	letsauth	1454340681	Notes from Friday (2016-29) Meet
Calvinized	anime	1454328988	Should I watch Chuuniby R
Cambijao	soccer	1454337801	League Roundup: Fre Ligue 1 - Match Week
Cantforgetmypassword	stopdrinking	1454343043	I just realized why it's hard for me to s
Cantinabandacoustic	swtor	1454338705	Trouble ordering on official s
Capablanca_INFINITY	HighSchoolWriters	1454325617	Prologue to a book writing, would apprec
CapnHighlands	Bedbugs	1454301997	Absolutely freaking about bed bugs, even
Capt_Ido_Nos	Guildwars2	1454344679	Use Cases for all combinati
CaptainAFAB	ftm	1454331430	Dysphoria, b dissatisfaction, who knc

		created_utc	t
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CarlProAV	videography	1454324058	I'd love to get feedback ProAV's youtube c
CastingMagilla	maker	1454346595	Casting for a Real MacGyver on a Major C
Castleofwolves64	furryrp	1454296388	[M4F/M playing F] [[Advanced] [Long terr
CasuallyClowning	Advice	1454302405	I Need Advice About Impending Future Abou
Catalytical_	SuggestALaptop	1454306620	[US] Looking fo lightweight school/gam
Catfish78uk	Fireteams	2908685394	[PS4] King's Fall - Hard Mo LF5M starting
Catsandradiobats	MECoOp	1454350846	No rewards after missio
Caughtx	GlobalOffensiveTrade	2908691368	[H] Karambit Fade 80/1 0,006 FV/M9 Marble
CaymanG	Debate	1454294723	There are NO 'elim' rou at Emory this Sun
CazwellNA	RantsFromRetail	1454302490	So fucking sick of gett put b
Ceannaire_Cogadh	Geosim	2908625205	[Event] Anti-Play Measures[Secret] U.S.A. I
CeeZeeBoy	canadagunsEE	4362879819	[WTS] Norinco M Shorty[WTS] Poly Te N
Ceipe	pokemongo	1454348342	Ingressing Pokemon
CelebRPFan	dirtypenpals	1454341655	M4F - Not A Typical Res (Fantasy
CelineHagbard	C_S_T	1454299917	Six Thinking Hat [6hats] i New Flair Type
Cenki	Fitness	1454288923	I stalled on strong because I didn't st
Centumviri	fo4	1454345362	The hidden value of walking arou
Ch3mee	PoliticalDiscussion	1454333156	Iowa Caucuses Predicti
ChainWay	GlobalOffensiveTrade	1454318933	[H] iBP Holo Shop! i Glock Fade & AK
ChaosRegiert	buildapc	1454335126	[Build ready] Overkill? W would you chang

		created_utc	t
author	subreddit		
Chapperion	XWingTMG	1454291209	Had a great night with new squa
Cheesey_Cakes	BabyBumps	1454329739	I need to make a sign for hou
Chewitt321	ftlgame	1454339809	New Multiplayer I
ChiBlock	magicTCG	1454297212	Ugin, Elder Dragons, i Tarkir (Vorthos Spe
ChiaPetBudz	wow	1454348056	Holy Pally Timewalk Dungeons
Chief_Iron	2007scape	2908679086	[Suggestion] Upgrad lower tier enchanted k
Chiiwa	RPGMaker	1454340780	Hello, question abou battle syst
Childree1	ffxivraf	5817294089	[Recruiting] [Any] Any ser Code: WSCR3BD
ChilliPC	buildapc	5817357667	[Troubleshooting] PC ke restarting[Troubl
Chimp-Spirit	entp	1454314903	How do we become the t that isn't among the
ChinPokoBlah11	Comcast	1454343235	Comcast is seriously PISSI me off and are
ChipperbrownXO	explainlikeimfive	2908579025	ELI5: If planets (earth) "pear shaped" -
Chippy569	lego	1454293814	This Week's Rebu Technics with a bonus
Chizbang	learnprogramming	1454341812	Switch to Lumen (Laravel Node
Chocobutts	raisedbynarcissists	1454318858	This is nowhere near worst thing she has
Chordata1	weddingplanning	1454354381	Upset over wedding c dra
Chris_Entropy	UnearthedArcana	1454340786	Designing a new "De Domain" for neutral c
Chris_plays_CoC	ClashOfClansRecruit	1454345833	[Recruiting] Eternal Hon Th 8+ New Clai
Chrispytoast123	MHOC	1454298559	Weekly Update
Chromi0	TheWitness	1454298657	(SOLVED) Help With De Obelisk, Third En
Chtorrr	Fantasy	1454349184	[crosspost] Marissa Me New York Times be

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author	subreddit		
	IAmA	1454349166	[crosspost] Marissa Me New York Times be
Chuck0612	buildapc	1454351630	[Build Ready] First F Please Advise - i5-4
Ciarkowski	GlobalOffensive	1454347394	New I
Cinnypoo	parrots	1454295111	Medicating amazo
Cinsare	borrow	2908679778	[REQ] (\$200) - (Toronto, C Canada) Will req
CitizenJac	ChicagoNWside	1454293585	Disabled Veterans Prope Tax Exempt
CitizenJosh	webdev	1454345292	Scheduling website: w would you u
CitrusGymnast	TWDRoadToSurvival	1454350594	Monthly Fact Recruiting/Advertis Threa
CivilatWork	buildapc	1454349329	[Build Ready] VR-Prep m Desktop (i7-4790
CjKvasir	relationships	1454310834	My [27 M] partner of 3 ye has broken up v
Cjyogi	computer_help	1454304713	I really messed someth
Cker11	thedivision	2908689716	Just a quick comment those concerned with
CleanUpGnome	UHCMatches	1454356684	Feb 05 20:00 UTC E Pierce's #43 - cTo5 - C
ClimateMom	marvelfans	2908619907	Character of the Week: Br BannerMatt Fra
ClintEastwood41	xboxone	1454356191	PSA: For old Hitman fa Alekhine's Gun com
ClosetYandere	RandomActsOfAB	2908680579	Daily Chat Thread Febru 1, 2016RAoAB Que
Clovericious	IronThronePowers	1454353754	[Event] A landing on fro shc
Cmdr_Nemo	SWGalaxyOfHeroes	2908625160	PVP: I keep getting attac first and mowec
CnosOriginality	asoiaf	1454291381	(Spoilers All) Any theories Jalabhor Xho
CoachingPikachu	FantasyWarTactics	1454310928	Looking for a bit of input heroes and tea

	created utc	t
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author	subreddit		
CptnPants	buildapc	2908657019	[Build Help] First T building a PC.[Build
CraterCat	TalesFromRetail	1454297924	"I Wasn't Born Yesterd
Crazedinqatar	relationships	1454347219	I (28F US citizen) live foreign country
Crazycatlady112	cancer	1454348422	Dads bladder cancer ke coming ba
CreamPieSexCouple	CreamPieSexCouple	4362994158	A picture from fucking ou nature [F+M]Ou
CrimOudin	TroveCreations	1454308606	((Spear)) Obsidian De
CrippleRipple	buildapc	1454356807	[Build Help] - Intel i5 4590 Xeon E3 123
Criskyfer	buildapc	1454321781	[Build Help] Building first for budget gam
Crobner	FinalFantasyExplorers	1454354226	Stacking Elemental Affini in Ability Mut
Croniclem	dirtypenpals	2908634878	27 M4F married looking seductive younger
Crooty	SuicideWatch	1454329952	Life just feels like a drain
CrossBack7	HFY	1454313266	[OC][30000]The Equaliz F
CrowmeatDPP	dirtypenpals	1454355166	[F4A] Uptown (C Downtown W
Cruecius	SWTORGuilds	1454354007	[The Ebon Hawk] <Dank Memes>; l I
CruelOptimist	FishMTG	1454306154	Best budget 1-drop: Judg Familiar or Cos
CrumbledFingers	antinatalism	1454339043	The worship delus
CrumblyButterMuffins	socialism	1454298046	Maine is going to hav ballot initiative tl
CrunchyGremlin	Warthunder	1454315096	how do you play the jun
Crushing-it	legaladvice	1454314764	(UT) Was just served pay for a student lo
CrypticTryptic	AskCulinary	1454305593	Can someone sugge good alternative to c
Cryptoviral	skyrimmods	1454353829	[Help] Random CT freezes and

		created_utc	t
author	subreddit		
Cubic_John	lifeofnorman	2908605650	Norman makes contact Norman accident (
Cuddle-King	relationships	1454300772	Non-Romantic: Me (19 Best Friend (19
Curelomcaprolite	exmormon	1454352571	Genetic bottlenecks i colonial viabi
Curvesandnerves	dirtypenpals	1454354584	[F4M] Curvy girl seeking v endowed partne
Cyarkos	PokemonTurfWars	1454286474	Phase 19 Ledger and Cap Locati
CyberWizardSC2	starcraft	1454338633	This weeks Tournaments NA- SCVRUSH (Tues
Cyber_Apocalypse	LightNovels	1454357912	[DISC] Reincarnated ~ Fr the upper world,
CyborgNonsense	Schizoid	1454305345	Vent: schizoid women i society's expectati
CzajaX	GlobalOffensiveTrade	1454349823	[Store] Some cheap kn some not so chea
Czekraft	seduction	1454310616	Skating on ice for a c
D9sinc	GiftofGames	1454309995	[Request] (Steam) Help travel 20 years ir
DBarrow15	squadup	1454293429	[SQD][Recruiting][Ser 34th Infantry Divis
DCDeacon	elderscrollslegends	1454297751	What are you playing w you're waiti
DJ4Bernie2016	SandersForPresident	1454298825	Last Call Into lo
DJAlex55	DotA2	1454350164	This is why I'm quitt Dc
DJ_Theo	Beatmatch	1454338577	DJ Name Quest
DMTsmoke	starcitizen	5817272727	Best weapons loadout Avenger?Questions re
DMeville	gameDevClassifieds	1454293450	[Paid] Cartoony Environment/concept ar
DONT_PM_YOUR_TITTIES	SandersForPresident	1454317699	We Choose to go to Mc
DPP4711	dirtypenpals	2908647600	[23M4A] Fuck my girlfrie (Other options i

author	subreddit	created_utc	t
DPPHannah	dirtypenpals	1454355007	F4F Fun with the beautiful Rose Ellen
DPPKate1	dirtypenpals	1454345708	F4A: Our own erotic science fiction
DPPTTheDoctor	DPPprofiles	1454336239	24M4A - I can play male and females
DPPxyz	dirtypenpals	2908651304	32 m4f -- The danger zone fantasizes of "on
DR1LLM4N	SuicideWatch	1454352424	Recently prescribed Xanax daily drinker, d
DRidder17	AskDocs	1454330903	Was on a weekend trip, sicker than I've
DTG_Bot	DestinyTheGame	2908634998	Daily Reset Thread [2016-01]Iron Banner Me
DUncle97	EliteDangerous	1454326971	Kill stealers: What
DaB0mbb	relationships	1454285525	[21/M] Having a tough time adjusting to the
DaEvil1	ELTP	1454346825	ELTP Next Sea
DaFazja	hotas	1454334969	"Captain Chair" for Elite Dangerous (Help need
DadAtH_me	homelab	1454349043	My BSD based
Daily_Dose13	aquaponics	1454355563	Newbie excited about getting started with an
Dan_B	ECE	1454351759	USB alternative to be support multiple ir
DancingPear	TryingForABaby	1454299857	Introduction ti
Danger1324	relationships	1454318265	My [15F] sister [13F] stole from me. Ag
Dangerbacon2222	Pegging	1454318704	First time Satur
DankWong	buildapc	1454316810	[Build Ready] AU\$1590 for B
DanknugzBlazeit420	DestinyTheGame	1454294997	Are all the solo players complaining about I
Dannnnv	relationships	1454317145	Me [39 M] with my girlfriend [33 F] 3 yrs tog
Dare89d87	Windows10TechSupport	1454356969	Constant graphics crash and HDD lockup. Any he
Dariusas	Anxiety	1454305961	Phantom pe

		created_utc	t
author	subreddit		
DarkGenesis327	MonsterMusume	2908602166	Liz Needs More Love!Are The Girls Otaki
DarkLorde117	RWBY	2908608408	[spoilers!] I Think We N to Agree on This
DarkRedTwist	NoFap	1454297142	I've never been so wr Thanks no
Dark_Element75	buildapc	1454292455	[Build Ready] Looking Advice regarding C
Darkbl00m	ketogains	1454353664	Blood
Darksider123	buildapc	1454347176	[Build Help] Using a sec graphics card fo
DarthTator	ClarkCountyAirsoft	1454288566	Welcome /r/ClarkCountyAir
Darth_Meatloaf	EDH	1454323363	Looking for help v cuts/sideboard swa
Daveism	sysadmin	1454350184	Exchange user continui business process l
DeaconX	thedivision	1454356323	The Open World Experience: Why the E
DeadManWONDER	pokemon	1454293345	Road to the 2 Anniversary: Pokemon Me
DearDarlingDearling	raisedbynarcissists	1454286311	Well, this is my st [TW:Abu
Dears1984	relationships	1454305324	Me [32 /F] with FWB/Crush[35/M] 1 y Fo
DecaffeinatedBarista	starbucks	1454295527	[US Licensed Store] H should my manager ap
Decap	TheWitness	1454297285	[Spoiler] Pulsing Laser: H requ
DeeSchro	buildapc	1454297925	[build ready] First t building, need some
Deepz3	h1z1	1454348988	The day Baseless sen (Nomad) are up, is
Degnir	csgotrade	1454319011	[H] Ak Redline ft (Titan K 2015 x4 and LG
Deityzilla	DestinyJournals	1454338796	Phoenix Rising Part II: Dead Don't Feel F

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author	subreddit		
DekuTrees_	cripplingalcoholism	1454319056	Severe stomach pain w drinki
Delsea	Guildwars2	1454347984	List of Guild Missions : can solo (for wee
DeltaDiezel	summonerschool	1454340948	A few observations made from Bronze V t
Delvakiir	DnD	1454344385	Is this magic item (
Delvin_McElroy	writing	1454354621	Now here is a story idea t I hope will ta
Demethius_J	Fantasy	1454336466	r/Fantasy Writer of The L Demethius Jack
DemonicSnow	CompetitiveEDH	1454349576	Need some help v mindset and deck choi
Denroll	galaxynote4	1454350606	Have an unlocked No from AT&T, but no
DerangedFrenzy	summonerschool	2908642209	Hey junglers, do pings assistance bother
DerivativeGhotiFish	dirtypenpals	1454307879	33M4F Who's up for sc benevolent sexis
Deskanar	magicTCG	1454332285	Missed line in the final GP Vancouv
Dessimation	funkopop	2908642763	Pop Stacking Contest &a Giveaway!(US) (H)
DevinXtreme	iphone	1454303020	What are some good er ap
Dexter87	IronThronePowers	1454284854	[Lore] The betrot
Dgameman1	asm	1454293874	I have many questions ab my assembly HLA t
Dhabu1999	Planetside	1454309158	The state of flight (contr in Planetsic
Di_Bello	Mywhynot	2908603703	Team of the Month: Phantom Troup
	whowouldwin	1454302795	Team of the Month: Phantom Tro
Dial_the_911	RWBY	1454304209	/r/BeaconAcademy: RWBY roleplay subredd
Diamano	hearthstone	1454346993	Learning Hearthstone: W Five. End of Seaso

author	subreddit	created_utc	t
DigitalSiren	FFXIVRECRUITMENT	1454286875	[Leviath <DelusionsofGrandeur& [LFM]
DimmiloSonoUn	getdisciplined	1454353318	[NeedAdvice]There is alm nobody that can
DinnerSonic	GearVR	1454316738	What store "Live De Units" can be used wit
Dirk_Breakiron	Knife_Swap	1454339193	[WTS] Price Drop! Br Division Black Wash (
DisneyMadeMeDolt	summonerschool	2908712312	Wayne tumble dama Support mains questic
DnDiene	metalworking	1454357483	Metalworkers, I need y help. What is this
Dobervelo	Dogtraining	1454338758	Doberman pulling on lea sort
Dobromr	pathofexile	2908626474	[HC][Trapper] For r league with the Acse
Doc_Ruby	thedivision	1454314308	My Dark Zone feedback hardcore Pv
Doctor-Swoosh	loseit	1454285654	My SV and various N over the first month
DoctorCheerio	gamemaker	1454298594	Problem with lengthdir "rotating cannon :
DoctorGluino	UnearthedArcana	1454342398	Artifacts: Shamanic Ite
DoctrVendetta	Trucks	1454300263	OBS Ford Powerstr FSV2A Install How
DoesMyUserNameSuck	SandersForPresident	1454333503	Thank /r/SandersForPresident the en
DogaLover	GiftofGames	2908595323	[Request] Sub Rosa[Offe bunch of gar
DogeyDogeDoge	buildapc	1454287234	[Build Ready] Is this look go
Dogmeat1210	DestinyTheGame	1454342596	Mutli-team playlist with r game ty
DohTellMehDat	GlobalOffensiveTrade	1454286012	[H] M4A1-S Cyrex (Mini Wear) [W] k
DontAskhowIknow	dirtypenpals	1454291991	25M4F - I Manipulated Into Breaking Up W
Dontsendmepics	personalfinance	1454338739	I'm a 22 year old that wa to start invest

		created_utc	t
author	subreddit		
DoritoScope	ClickerHeroes	2908691819	1 Million HS opti ascensionsManual input
Doron_B	buildapc	1454312716	[Build Help] Gaming: M settings, 1080p, 45f
Dpp_dominant_teacher	dirtypenpals	1454356069	M4F Professor and stud
Dr--Death	syriancivilwar	2908689725	Northern Alep OffensiveSyria: Syrian Ar
Dr-Fruit	Unity3D	1454335940	Networking question / ic
DrAAE	GlobalOffensiveTrade	4363006201	[H] 4xHolo Kato14 St AK serpent ft [W] I
DrTestBender	linuxmint	1454300797	I'm running 17.3 Cinnan and I'm getting a
DrTxn	exmormon	1454299072	Divorce crazy TBM be witness of divorcing
Dr_Hydra	thedivision	1454297651	What about a Karma syste
Dr_Zoidburger	Pins4Skins	1454312986	[H] Guardian Elite / Dust 3 / Mirage - P
Dracious	dotaimba	1454289716	Imba Beastmaster / Lacking? Anyone had an
DradDog0419	personalfinance	1454291250	Tax Fr
DragonDeadite	TalesFromRetail	1454341448	"I'm Afraid I Can't Help Yo
DragonSlaayer	summonerschool	1454351198	Can someone explain Da to r
Dragonheart79	FFXIVRECRUITMENT	1454330289	[FC][LFM][MC][Cerbe Order of Bahamut is
DrakeH44	GlobalOffensiveTrade	1454298597	[Store] 47x Knives @ 8 Marketp
Draqur	legaladvice	1454347207	Property damage by dr driver, his insuran
DrawerFullOfDicks	rapecounseling	1454300328	It's my birthday toda finally told my mor
Draws-Your-Request	juststart	1454341985	Hiring VA's and runn multiple websi
Dreyri	runescape	1454349963	How disassembling wc
Drovid	NoFap	2908662526	Insider tips for success.W the fuck do yo
DrowSpectre	RWBYOC	2908683365	Team LVDR - Roux BriarTe LVDR - Daegan R

		created_utc	t
author	subreddit		
Droyk	Enhancement	1454306762	[Discussion] having prob with Styleshe
	RESissues	1454307564	[Could be a bug] Hav problem with Styleshe
DrunkParty_Girl	dirtypenpals	1454304881	F4M - Dat
Drunk_Conlangs	conlangs	1454301366	I drunked a conla
DualKnives	thedivision	1454304187	Just got spawn killed for ho
DuckTalesLOL	vegas	1454350336	My experience in Vegas the first time fo
DudePickle	thedivision	1454346495	My First Impressions of Divis
DuleHillSexDream	GradSchool	1454326897	Venting/advice appreciat nightmar
Dunadan019	FFRecordKeeper	1454342502	Accessories advice/wha
	PostPreview	1454341855	
Dundus	TheBestAlog	23268990520	I killed 3 K'ril Tsutsarot defeated the C
DuneManta	techsupport	1454301779	Galaxy S4 randomly sh down and won't res
Dungbeetledonut	collegeinfo geek	1454347981	How I arrange my own es assignment files c
Dylando	pettyrevenge	1454320246	Flame me while I'm abou leave a game? Le
Dylarno	UHCMatches	1454314717	Feb 01 09:00 UTC A Dylarno's #104 - FFA -
DynamicNewAlgorithms	FATERpg	1454307546	After action reports I've b writing for m
EBuni	CFB	1454352022	[FCS] Post-Season Recap - Charleston South
	fcs	1454352051	Post-Season Recap: # Charleston South
EDUARDOMOLINA	podemos	1454356389	JUAN CARLOS MONED IMPOTENCIA.- Público.- I
EFECTER	GlobalOffensiveTrade	1454344888	[H] Skins with Kato 2 stickers [W] K
	csgotrade	1454344898	[H] Skins with Kato 2 stickers [W] K
EG_iMaple	JRPG	1454336180	JRPGs and Waifu Pander

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EVChargeSolutions	BMWi3	1454339467	Hello fr EVChargeSolutions.co Here to he
	KiaEV	1454338606	Hello fr EVChargeSolutions.co Here to he
Ebessan	SquaredCircle	1454292431	Man vs. Woman Matche Lucha Undergrou
Ecksacutioner	DestinyTheGame	1454340082	Because of only match wins on a card, Tr
EclecticArtificer	pcmasterrace	1454355626	So I bought the Logit G930s, have I made
Edemardil	ElderScrolls	1454301733	Just a random Elder Scr memory (Possible
Editorgirl2617	relationships	1454296345	[30/F] tired of pas: aggressive & ext
EgnarmythPro	luchaunderground	1454304322	Is it really wrong for me feel bitter? [Re
EgoSaber	exmuslim	1454336785	TIL Why Muhammad Wai Three Ye
EightsOfClubs	heroesofthestorm	1454348081	[Blizz Pls] A suggestion feedback on repc
EknimEagle	leagueoflegends	1454325993	Lacked 7 RP for Me Kha'Zix, wrote a poem
ElFabio	dirtypenpals	1454306334	M4F A tale of two lov (vanilla, romar
El_Nahual	Cuckold	1454301581	Shakespeare the c
ElectricPaperMajig	CruciblePlaybook	1454351604	Destiny break-up sa video with detailed c
ElenaSzeliga	recipes	1454326020	Mussels in White W Sa
Elfire	summonerschool	1454351172	A Question Regarding Jur
Elijah_Abels	exmormon	4362899573	Crazy Mormon Bisl Sayings...(long)Interview
ElitistBlack	Warframe	1454341965	[Suggestion] Allow play to solo extract f
Elivaras	LeagueofAngelsMobile	1454284837	A Guide to Emble
Elleseven1967	relationships	2908700832	Me [48F] with my boyfri [56M] of a year ar

		created_utc	t
author	subreddit		
ElmerJ	fantasybaseball	1454346186	Real 'Fant Baseball....Dream Team S
Elmidra	fantasywriters	1454328090	Looking for dedicated k readers for my da
Emberhair	bloodborne	1454342778	The Eldritch Trutl Collection of Bloodborn
Emilia23	Advice	1454304140	My bf is a bit older than and thinks I'm r
	self	1454304869	My BF who is bit thinks should wait to ha
EmilioSleeves	xxfitness	1454285737	Currently struggling accept that I need t
Emotional_Ewok	porcupinetree	4363005090	Song Discussion: Week (Open Car)Poll: Wee
EndTagScorpion	Dermatology	1454336965	Trying to help my fian heal from severe a
Energy_Turtle	Pmsforsale	1454303445	WTS - Bargains for everyc Dollars, Einste
EnlightenedViking	asktrp	1454353470	Fucked me like she lo me. Part of the man
EnochSamways	nosleep	1454316543	I Looked Into Its E
EnragedTurtle	clothdiaps	1454336653	Where to find inexpen: non-microfiber inse
Envimea	mcservers	1454292329	Tempus Realms [SMP][P {Towny}{McMMO}{Jobs}{
EpicDildo	Homebrewing	1454292017	When you forget hone fermenta
EpicEnder99	TrueFilm	1454342590	Am I the only person v didn't feel satisfie
Epic_MC	streetwear	2908587234	Spotify Playlist... Let's try againID o
ErinEggplant	wow	1454315735	What is up with pec levelling in this gar
ErnestoGrimes	arduino	1454302976	I could use a bit a help v my ske
Erotaku12493	ModernMagic	1454314052	Thoughts on my L Control I
Erotic_espionage	eroticaauthors	1454346910	Trying my hi
Eroticdreamz	offmychest	1454292397	Playing Pret
Escaflowne	escafrisky	1454291350	requests for my voice

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author	subreddit		
Eshtan	conlangs	1454311095	Generating Grammar fr Scre
Estabon	smashbros	1454332822	Corrin and Bayon damage out
Etrae	whowouldwin	1454323567	Banner Fight - Febru
Eunovation	PuzzleAndDragons	1454284990	[Question] Should I Res My Gar
EurasianTiger	hapas	4362982924	The last two Bond mov Skyfall and Spectr
EvenFlowX93	SquaredCircle	1454294673	Let's try to get Ambros the Wrestlemania
EvilAngel00	chrome	1454334670	Full screen video cras Chrome on Windows
Ex0HanZ	Pins4Skins	1454351190	[H] All kinds of pi Reasonable prices
ExquisiteInquisition	depression	1454340685	My sister accidentally t me how to kill my
Eyes_on_fire_0323	nosleep	1454312359	Always Drink Respons
FAWmod	ForeverAloneWomen	2908644696	[Meta] New addition to moderation team[F
FLICK_MY_BEAN	legaladvice	1454354256	[NJ] Best approach evicting a potential
FLYBOY611	netsecstudents	1454350424	FLYBOY611's Guide Starting Out in Compute
FOOKCONSOLE	zenfone2	1454338728	Phone D
Fa11enAngelIV	XenobladeChroniclesX	1454293874	Lots of ponderings questions about end ga
FabioTheBrain	AzzuroRedoubt	1454314956	"I think I've figured out w happened that c
FactArchitect	personalfinance	1454310688	The story of FactArch and requesting h
FaeriaTester	faeriatestsubreddit	1454356111	Bacon ipsum dolor amet j tongue boudin, st
FahyZ	pcmasterrace	1454320508	Dual Screen colour match issu
Faithless996	exmuslim	1454348682	Concept of a sa
Falklander	dirtypenpals	1454353530	23 [M4F] A technical ma
FallenAngellI	AskSF	1454320771	Nerdy/Geeky/"Gay" thing do between Febru

		created_utc	t
author	subreddit		
	SFFood	1454321376	Looking for good seafood and Asian cuis
FallenRenegad3	empirepowers	2908585808	[EVENT] Maml Recruitment[EVE Conquest
Familyman239	Addons4Kodi	1454348833	Kodi AFTV-Best update build w/out Genisis F
FantasyDuellist	ftlgame	4362955596	4 sectors of ShivanWhat the boss?Tales
FantasyNBAMod	fantasybball	1454329842	Week 15 Helpful Content Feb - 7 F
Fapforfapssake2	dirtypenpals	1454356816	[M4F] Holy Roleplay Batr
FargusFaustmeister	canadaguns	2908605718	No matter how bad groups are, she always n
Fascion	cordcutters	1454333307	Indoor antenna inqu
FaurenFurus	GlobalOffensiveTrade	1454351983	[H] 2x FT Fire Serpent (C and 0.16) [W] K
Fazermint	PowerMetal	1454332836	This Week in Power Metal Releases (02/01 - 02/
FearlessFixxer	exmormon	4363028014	When discussing fallibility/infallibility wi
Feasinde	tolkienfans	1454301267	[Language]What do you know about /r/ in Queny
FeatherArm	legaladvice	1454353655	Apartment has a lease violation different dates tl
Feezec	xcom2mods	2908601109	Imperial Guard CommissarAir Support
Fennex_	Paladins	1454343922	Looking to get a team go
FerralOne	gamedev	1454289736	Important or useful Algorithms and M conce
FerrisTriangle	SandersForPresident	1454286344	Comparing project turnout to rally attendance
Ferry83	FutureFight	1454335479	A bit of Crystal guida
FetusMonkey33	buildapcforme	1454298013	Strictly Streaming
Figs232	snes	1454330635	What's your collection strate
Finn_Finite	SchoolIdolFestival	1454303257	[Information] JP Med updated breakdowns

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Finseed	heroesofthestorm	1454339604	Will we ever see an Azoth themed Li-Ming sl
Firstyearten99	GeneralTaxAdvice	1454344762	Should I have already filled out a 10-99 if
	personalfinance	1454345985	(USA) Should I have already filled out a 10-9
FishKisses	weddingplanning	1454330722	How to cope living apart from FH up to and a
FishoD	Vermintide	1454348700	Why do people care about loot so much? It's
Flamingtomato	witcher	1454308115	Select starting Gwent d
FlandersNed	HaloOnline	1454332493	The Chronicle of Mista (one of the better
Flarebrass	Starcitizen_trades	1454356124	[WTS] Many LTI Ships at c + MC
Flatoftheblade	CK2GameOfthrones	2908572908	Hilarious bugUh...what h
	gaming	1454316465	Video games v ambiguous moral choic
Fletchyo	GlobalOffensiveTrade	4362967712	[H] Awp dlore FN 0. panic sell.. [W] 567
Flewtea	breakingmom	1454298657	Welp, that's my good kai for this y
Flixxxxo	starcraft	1454306784	For the people who want ladder more but ca
FloridaFriend9000	r4r	1454291602	23 [M4F] Central Florid Quirky book lover
FlowerpotJason	GlobalOffensiveTrade	1454349672	[H] 27 CSGO Keys [W] USD paypal e
Flying_Panda	DnDBehindTheScreen	1454356690	Dragon's Nightm
FoXx1337x2	GlobalOffensiveTrade	2908626362	[Store] Doppler, doppler, doppler. k
FoamNinja	battlefield3	1454350631	The hacking is to r
FocusTaiwan	taiwan	1454316493	Death and Destruction fr Snow Storm Domina
FollowDurdenHS	hearthstone	1454286738	Give Me Your Data Publish the Rep
FollowMyWords	dirtypenpals	1454353165	[M4F] How to Turn Your into a Good Girl
Foom92	MenWithToys	1454357292	Beginr

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author	subreddit		
ForScale	javaScriptStudyGroup	1454348898	[Week 3] Focus: Recurs
FortBass	BoomBeach	1454357210	Mega Crab added a r dimension to the ga
FortHamsterdam	army	1454337164	February 2016 /r/A Professional Developme
Fortinbraz	ecrpoker	1454333787	ECR Poker - January 31 Res
Foshkey	ffxiv	1454340116	Just played the first th 24-man raids las
Fossana	poker	1454312261	I'm not meeting my pc goals, and I'm wond
FouFromage	forge	1454339228	[Completed] Shoots_Red Counter Strike Rem
FoxyRin	pcmasterrace	1454337434	Winloader & Simila How (un)safe are th
Fozeri	GlobalOffensiveTrade	2908626532	[H] 0.19fv AWP Asiimov F BTA 0.25X AWP A
FranMirren	relationship_advice	1454307099	My bf (28/m) is verbally i emotionally abu:
Franszon	OnePiece	2908666187	[SPOILER] This was actu predicted 2 year
Frau_Aeron	dirtypenpals	1454311321	25 F4A- The Kitsune's Bl
FreddeCheese	Fantasy	1454354885	Recommendation fo former reader of Fantas
Freeek323	TheWitness	1454341346	[SPOILERS] Did anyone c use math to solve
Friendwontleave	relationships	1454288979	Me [23F] with friend [2 that constantly in
Frikster	Laptop	1454296194	AH-IPS vs IPS; i7-4790 vs 6700; DDR3L-160
FrozenLegend99	youtubers	1454351539	[Review Video] New Ho Gaming Channel! Le
FrozenSundae	amazon	1454312510	Lost Access to Acco & Orders - Any exp
FultonExtract	playrustservers	1454339140	New EU(DE/EN) 200 server - ballerbude1
FumuR	Pathfinder_RPG	1454346941	Need Help with Speec Combat/Adventure F
Funkyokra	skiing	1454356585	Older but new wom lange bc

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Furious_Sleeper	microcontrollers	1454340699	A little help needed with PIC16F877a proje
Fus_ro_dah_lady	xxketo	1454342916	How am I STILL hungover need your stori
FynnCobb	Treknobabble	1454336603	PRIORITY ONE PODC, 256 THE (BIG) AP
GALACTIC_LAW07	IronThronePowers	1454350779	[EVENT] Don't Shoot Hei
GMCsonoma4x4	NoFap	1454298689	why only quit one addict lets make it t
GREKOE	GlobalOffensiveTrade	1454352114	[Store] big premium sto kara ruby, kara
GSG1901	sabres	1454308544	O'Reilly at the /
Galbalbator	fantasywriters	1454303598	[Critique] A Review of Magic Syst
Galokot	galokot	2908711094	The Misadventures of ' and AsmoWill and As
Gamegeneral	gildrune	2908685204	[Recap] Session 11(Every aboard the murde
Gamer5Lyfe	techsupport	1454344088	How Do I Get Rid of a Vir
Gamertag1236547	feedthebeast	2908652278	Anyone use mcdungo with modded minecr A
GandalfTheyGay	thedivision	4362985241	Can we add the option inspect people?Anyon
Ganman3	buildapc	1454310944	Working on a build- a li overwhelmed [B
Gargalensomething	GameofThronesRP	1454301254	A Wise Man From the Nc
Garrod_Ran	Gunpla	1454297715	What are the available 1/ scale and SD Gur
Gate4043	Steam	1454344423	DLC for Lego Marv Avengers not instal
GattDayum2	tifu	1454288674	TIFU by being an optimi third-gra
GayWarden	FanTheories	1454298979	[Narnia/The Magicia Fillory and Narnia co
	brakebills	1454302635	[Books] Fillory and Na coexist. (x-post
Gear5th	Anki	1454294719	[Guide] How to Anki Ma the right v

		created_utc	t
author	subreddit		
GeneralPsycho5	cigars	1454293072	[Review] Cama Connect
General_Tsuki	RSChronicle	1454339794	5 Things to do after the c wipe for newcon
Gentle_Beard	bjj	1454295354	Would you apolog
GeoSparker	GBr4r	1454292000	22 [M4F] SW Midlan Anywhere - Pros and C
Ghastily	tf2	1454351816	Valve can you please demoknic
Ghillieintehmist101	TrueDoTA2	1454327102	Request : Help with Em Sp
Gibusmann	The_Donald	10180424320	Today is the dawn of TrumpeningA quick re
Gin_chan	StrawHatRPG	2908707043	Stats for 1/1/ 1/2/16Ganseki Pai Capta
Ginaguo411	3FVAPE	1454292632	3FVape Daily New (1601
GiraffeNjah	relationships	1454348489	Me [28F] with my l boyfriend [30M] of one
GlutenTag	legaladvice	1454304876	Sold on Craigslist bu demands refund. Adv
GnarlyBellyButton87	starcraft	1454296958	Satellites are used to l around the battl
GoatMr	dirtypenpals	1454303153	33 [M4F] Horny Charts : Naughty Graphs: Li
GodKeeper	Fallout	1454318851	(SPOILERS) Fallout 4 Anal
GoldenJermbag	DestinyTheGame	1454326080	I've finally had my first t high sodium mo
GoneWriting	travel	1454307997	Skip Cesky Krumlov for m time in Berlin? !
GornoP	Anger	1454352756	The "E-Wor
	Anxiety	1454353369	The "E-Word"... (x/post fr /r/Anx
GottaRiskIt	arduino	1454343801	Creating a SUPER-BUDG wi
Gourgeistguy	bladeandsoul	1454342829	Do you personally beli this game will ge
Gradepuxi22	Trove	1454349189	A concept for post-rad cont
GrantTurismo	lifeofnorman	1454335368	Norman and GrantTuris

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author	subreddit		
GraphicTitan01	RWBY	2908671150	Community Design Ch Spectacular Character
Gravlox15	selfpublish	1454330700	[META] What would guys think about an AM
Griffin_Throwaway	dirtypenpals	1454298312	22 [M4F] Rule 34. Who invented it deserv
Grimsrasatoas	Guitar	1454327043	I'm finally beginning to able to play the
GrizzlyS	NoFapChristians	1454287263	1 week
Guardax	nfl	1454334489	50 Super Bowls in 50 D Super Bowl X
Guitarface1143	RandomActsOfBlowJob	1454341192	[m4f] Netflix and Chi #aus
GunSizeMatter	buildapc	2908582218	Quick GPU Question ab 144hz 1440pFirst Sa
Gundel221	GlobalOffensiveTrade	5817386029	[STORE] 14 ibp/titan h 2k14 Katowice weapc
GurrcatSWG	swg	1454349529	New SWG NGE Proq
GuyMontagz	spikes	1454293415	[Modern] SCG Colum Modern Classic Top 16 I
GuyNoirPI	asoiafcirclejerk	1454295807	[Crow Business] Need Changes to the Spoile
GuyWithoutABeard	PickAnAndroidForMe	1454325412	Stuck Between two cho
Guy_Tell	Bitcoin	1454321336	My thoughts on what co happen if Classic
Gym_Leader_Erika	SVExchange	1454355718	Box Clearance Semi-FFA Giveaway Part 3:
GypsyRonin	malaysia	1454317123	Getting Internet Servic Putraj
GyrokCarns	OutreachHPG	1454355234	ELI5: MWO and position
HDE01	neuro	1454305392	Does anyone know w feedback loop is respo
HIGHlifequad	multirotor	1454343086	Help a noob
HMSBounty-A	DaystromInstitute	1454342881	Is a Ba'ku lifestyle effective Borg deterre
HMS_Regulator	modelmakers	1454325655	Is it normal for Trump model instructor
HSAHughes	DvZ	1454298462	New Ranger H (Suggesti

		created_utc	t
author	subreddit		
HabsGirl	PersonalFinanceCanada	1454291806	Savings for a c
HakunaVajayjay	movies	1454326370	The Hateful Eight aka Boring Three (Ho
Half-Invented	Trichsters	1454348429	Hypno
HalfChubby	pokemon	1454356552	Favorite Pokemon Resu
Halfayear	relationships	1454296329	Me [20F] and bf [2 recovering from emotio
Hammeredmantis	relationships	1454286801	I (26m) am just...bored at point with m
HamsterHawk	GlobalOffensiveTrade	2908646077	[Store] Blue butt AK CH EnVyUs Masterpi
HankDeLaHill	findapath	1454316205	24/M searching for a car in a rural a
Hannahlebanana	DogCare	1454300219	Should I sell my dog to k my husband happ
Hard_To_Let_Go	offmychest	1454308965	I wish I could apologize every
HardandLost	runescape	1454307582	Skill Reset Clarificat
HardlyDangerous	Eve	1454350677	Thoughts on EVE after first y
HashtagButterfinger	BDSMpersonals	1454293995	35 [M4F] Romantic clueless sub in search
Hate_Consulting	consulting	1454356135	Been in Consulting two ye need exit opti
HayzerUnlimited	PersonalFinanceCanada	1454349546	Might require a loan continue schoo
HazMatt082	offmychest	1454334312	I can't fathom how pec make such signific
HeartbreakHarbringer	dirtypenpals	2908651677	27F4A - It's a slow day.27 - Here's the d
HeartlessOne123	GlobalOffensiveTrade	5817308128	[Store] Kara TT & Sc Mid-Low Tier Stu
Heartoplease	dogs	1454300012	[Help] [Discussion] My c has bacterial hep
Heavensguard	thedivision	1454309024	Interesting game within ga
HelloUrineforatreat	AgeplayPenPals	1454322076	M4F Fear Fac Roleplay...updated a k

		created_utc	t
author	subreddit		
Hello_There2	worldroleplay	7271767864	[NEWS] Self-Defense[NE Germany drops sanc
HeloRising	rpg	1454355978	What's a good way handle henchm
HendrixDae	learnprogramming	2908659223	[C] Update! Reworked cc still having issu
HenryKofalin_2113	casualiaama	1454348695	My name is Henry Kofal am (or was) a co
Hentie_34	dirtypenpals	1454291643	20 M4F who doesn't l some fun rule
Her_Royal_Bitch	raisedbynarcissists	1454302762	The realization that "hor is worst trig
Heraldry_contests	heraldry	1454302472	February Arms Des Con
Hermes13	heroesofthestorm	1454353473	Weekly Battlegro Discussion : Dragon S
Hero_of_Seals	fireemblem	1454354678	Sharing my thoughts a finishing Radiant I
HerpaDerpaShmerpadin	vandwellers	1454304855	Tips for staying ou trouble when living
Hezekiah_Winter	blacksharkcult	1454324443	Fleet Commander Trair thread - Doctrine co
Hi_islander	Hi_Islander	39266336509	48000000XP in AttackI ki 59 General Graa
HiddenInquisitor	atheism	1454306750	Pentecostal church grow [R:
HiddenMosquito	spaceengineers	1454348519	Shrine of Cl
Higgzi	freedonuts	1454311579	[GIVE] I am giving Dor 🍪 , Cash 💰 , Tickets 🎟
HildasAster	Skyforge	1454315885	PHOENIX ORDE Recruit
Hillsmills	OSVR	2908585206	Isn't Anyone Using Tr and/or Trinus with
HiltoRagni	cigars	1454325001	I'm new here + my week smoke - Hoyo de Mo
Hobo_on_a_Stick	relationships	2908610479	My (18m) mom (58f) volatile control fre
HockRivers	Advice	1454292832	(24/M)Need a dose of rea and some perspe

		created_utc	t
author	subreddit		
HolidayNick	relationships	2908572114	Me 21 Male and girlfriend 4 years need he
Holo_of_Yoitsu	anime	7271724099	[Spoilers] Diamond no / Second Season - E
HomicidalHotdog	DnDBehindTheScreen	4363037627	Rules, Mechanics, and r DM Questions Megat
HonkersTim	EliteOne	1454328071	Introduc
Hopp3r	talesfromtechsupport	1454338423	Simple proble
Hotbyassociation	AIAnon	1454330887	A little
HowDoYouLikeMyName	GlobalOffensiveTrade	1454289395	[H] 2 Kato'14 USP-S's Stainless, Dark W
HowlingWolven	PhantomForces	7271684640	On machine gu [Balancing]My support p
HoyaSaxons	loseit	1454292395	NSV -- I was told I ins some
Hpanduh	NoFap	1454311275	Trying to avoid the bir
Huddy1299	GlobalOffensiveTrade	4362995207	[H] That Insane Double Holo'd Redline [V
Humminglady	CasualPokemonTrades	2908592355	Shiny Flair: January Winr & February \
Hurricane808	GlobalOffensiveTrade	2908644485	[PC] Falchion Blue gem Falchion Blue g
Huskynoob	Dogtraining	1454335431	Recently got my dream i (Sinerian Husky) b
Hwadu	zen	1454336457	Koan of the Month: T Yen's Eyebro
HybridMatty	photography	1454307397	Nikon 18-55 VR II vs 18 E
I-iz-Ratty	pcmasterrace	1454324178	Helping an old school fri reaso
IAmFrankCamberlain	belgium	1454340069	I am Frank Camberlain me anythi
ICantAwpForShit	starcitizen	1454333661	A discussion about gra on st
IEatGermanChocolate	emotionalabuse	1454323283	The tables have turr
ILiftVid	Fitness	1454330777	Form check SL 5x5 work A (low bar squat, l
IM_A_DOCTOR	BabyBumps	1454347216	Food poisoning (I thinl the wc

		created_utc	t
author	subreddit		
IRainstorm	noveltranslations	1454322415	[CN] Zhu Xian - Boo Chapter
ITS_JUST_A_WANK_BRO	The_Donald	1454345945	Funny thing happened party on Fri
IVlattEndureFort	DIY	1454299948	Is it possible to make/b glass for window
IWillNotLie	Grimdawn	1454298754	(Spoilers) About the plo Act
I_AM_ALWAYS_ANGRY	EthereumSpam	1454341944	[SPAM] Hello - /u/synah (Hilarious. "I'r
I_Am_An_Audio_Noob	hometheater	1454349284	What amp should I get? many to choose fro
I_DID_LSD_ON_A_PLANE	2016Elections	1454301932	Why is it Trump vs C Clinton vs Sanders :
I_Luv_Oreo	hearthstone	1454285659	Looking for Feedback on first Deck (Newbi
I_askthequestions	corruptscience	1454350840	Unbelievably bad science equation for det
I_tinerant	Bonsai	1454351745	Spring is Nigh - Any Adv on 3 Proje
laerin	nosleep	1454315607	I'm ready n
IanAttak	DnD	1454331314	First time DM... And players want me
Icanneverremembermy	RedPillWomen	1454343849	Handling conflict betw your long term SO :
Iceburnchaos	disability	1454308312	What steps can I ta
IcemanLoki	GlobalOffensiveTrade	1454345489	[PC] 60% Blue! C Hardened Flip Knife (V
Ichigowins	Ichigowins	1454287635	POST WEEK 3 P RANKIN
Ichiorochi	DotA2	1454327797	I did not see anyone hav done the math fo
Icronics	hcteams	1454289207	Map X: Spawn Fights : "Logging"
Idahosquirrel	IDontWorkHereLady	1454351515	I'm telling your manag
Identimental	Fitness	1454329647	Light chest work on shoul day of a full-bc
Ifette	flying	1454351032	Most reasonable way to CFI c

		created_utc	t
author	subreddit		
IhaveaGFnow	relationships	2908703398	When and how do I [28f] say "I love you" to n
Ijustcantsleep	relationship_advice	1454293353	My [27f] SO [30m], porn & masturbation. H
Ilan321	dating_advice	1454343148	First timer he
IIlManTheFlashlight	relationships	1454297399	My (21F) boyfriend's si (26F) always exp
IlliterateEngineer	tipofmyjoystick	1454297636	[PC][Mid 90's] Point & Click from futurist
Illsonmedia	DragonsDogma	1454285466	Should I respect vocati
Illum503	SubredditDrama	1454323747	Opposites attract r/electromagne
ImBae	kohi	1454293642	Help with PC Problem (F
ImBasicallyTrash	hearthstone	1454326402	Issue with Itunes & payment system for Hea
ImRickyBobby	opiates	1454290833	Heroin not affecting much? Daily Kratom U
Im_Awesome56	nosleep	1454302605	I wish it had just beer obnoxious dr
Im_an_Owl	HomeworkHelp	1454356352	[Uni Differential Equatic Solving for t
Im_so_mad_bro	Swissosugame	1454339977	[Meta] Subreddit Chang New Rules and *hop
Imallagog	buildapcforme	1454287467	For my dad tech-wary c
Imipolex42	ProsePorn	1454293701	Gravity's Rainbow - Thor Pyncl
ImperfectXIII	listofponies	10180430816	Earth, m, coat:pale gre mane:grayish tea
Impetuouspeach	relationships	1454346261	My best friend[19 F] br up with my other l
ImpossibleSuffering	offmychest	1454314111	I feel like my so interaction enjoyment n
Incom_T65	ketorecipes	1454340373	Thoughts on my upcom *real* pasta dough e
Ineedanameplss	ContestOfChampionsLFG	1454316024	1.2 million alliance requ free-loaders, c
Ineedhelp0091	DeadBedrooms	1454319220	attempted sex - he admit to erectile proble

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author	subreddit		
InfernoSquirrl	DotA2	1454349125	I'm 1k and my team dr me dc
Infinisecond	personalfinance	1454357808	Looking for suggestions how to appeal to
InfiniteEclipse	GlobalOffensiveTrade	7271696036	[H] Playskins! (Ak-47 Vul WW with Katowic
IngloreuzBstrd	skyrimrequiem	1454297945	[RP] Scales and Sca
InoPony	MLPLounge	1454315035	Tamest Midlife Crisis E
InsaneOne8977	GiftofGames	1454288421	[Request] Guild War Heart of The
IntelligentCommenter	nfl	1454356586	Officiating apocalypse - w would happen i
InterRail	bladeandsoul	1454346215	Let's bring up End-ga public party iss
InterestedIndividual	NoFap	1454309348	Addicted? Turned g
Iridar51	Planetside	1454288734	[GUIDE] Bad Advices: How attack someone's
IrishLaaaaaaaaad	WrestleWithThePackage	2908644336	And the WINNER PACKAGE OF THE YEAR 2
IrishinManhattan617	DIY	1454348603	Help: Converting sir switch to a switch/o
Iron_Rogue	Essays	1454302428	Need to ask a clarify quest
Irresponsibility	90daysgoal	1454288727	[Daily Goal][Check In] Day - February 1, 2
IslamMostafa	techsupport	1454321547	Loud Laptop
Islandhopper401	stratux	1454350497	Traffic Altitude Quest
ItCameFromTheSkyBeLo	SandersForPresident	1454352384	Regardless of what happ today in lov
ItRhymesWithCrash	yugioh	1454313484	[R/F]Blue-Eyes for cas f
ItsKindaFunnyBecause	tipofmypenis	1454303723	Homemade sho masturbation vi
IwontTryAnotherName	proED	1454345450	I cannot belie
Izzenw	TheRedPill	1454327334	Brazilian Crysis. Women almost 70% on Br
Izzy10235	asktransgender	1454305055	Planning surgery at sc point. Any of you h

		created_utc	t
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J-u-l	csmapmakers	1454349156	Custom so
J3Tisgod	respectthreads	1454285100	Respect Goten (Dragon
JBB1986	Supernatural	1454303177	[Spoilers] Two thin noti
JENNrationalization	r4rmidwest	1454343288	23 [F4M] Iowa/Midw How do you trace a sc
JES2140	buildapcforme	1454296475	First build - Budget \$1
JGPH	AndroidQuestions	1454293599	New to Android and n h
JRHelgeson	networking	1454348646	Popular site for IP thi landscape, Norse C
Jaamaaal	GlobalOffensiveTrade	5817307183	[H] Double-Sided Bla Gem 90% PS+Sweet Adc
JacP123	circlejerk	1454354339	The Fine Bros. annou their intention to si
JackBlackJackWhite	dating_advice	1454333052	Does It Sound Like This Likes M
Jackattackie	geocaching	1454344215	Help adapting geocach for my grade
Jackinthebox13	dirtypenpals	2908626580	19M4F- Dirty DND!19M Dirty DnD, No Expirati
Jacksbackbaby	rpg	1454355366	DMs and GMs of Redc implore your assista
Jafoob	relationships	1454284814	Me [24 M] with my Girlfri [22 F] on milit
Jak3theD0G	yugioh	1454285450	Nekroz
JakeVanderArkWriter	Spanishhelp	1454291071	I'm about to publish fourth novel, and I
Jakecoop2822	GlobalOffensiveTrade	1454293253	[STORE] Bayo TT Katowice 2014 sk @lov
James812	Dell	1454340841	Dell Ordering I
Jammerben87	Grimdawn	1454326531	Questions from a new pla
JarnabyBones	SandersForPresident	1454296004	It's all about the Su Delega
Jarunik	kundalini	1454352929	The Basics of a Balan Lifes
Jason_Webb	ADHD	1454352975	Thoughts on Waking Getting work done, an

		created_utc	t
author	subreddit		
Jataki	PrimeiraLiga	2908657134	Resumo da última sem no r/primeiraligaKuc
Jay211	GlobalOffensiveTrade	2908589181	[H] Vulcan MW, Kara Awesome Corner and Bay
JayConz	gallifrey	1454331244	An Ode to Netflix, and w think that takir
JayOhhGames	Unity3D	1454308747	Looking for feedback on of my asset pa
JazzberryRose	kpop	1454288957	When did you realiz group had become your
JeCsGirl	harrypotter	1454347270	February Extra Crec Astronomy [St
Jeanca92Panda	unitedkingdom	1454303620	Tom hardy, Legend. segm
Jeff-Stelling	unitedkingdom	1454356081	Prestatyn photobomb hc owner complains no
Jelloriver	NarutoFanfiction	1454308131	I can't remember the na of this fanfictior
Jellyfishpuff	learnpython	1454353207	Help with my lo
Jelre	GlobalOffensiveTrade	1454286749	[H] ST Karambit Doppler ST Auamarine, AW
JerimothHill	mazda	1454334347	2015 Mazda 3 doors loc when they shouldn
JerusalEmAll	cigars	1454286790	Celebrating Day 31 – l Privada No.9 Flying
Jessicaglitter88	legaladvice	1454350627	Filing a wage claim aga former employe
JezquetTheKhajiit	Jezopolis	1454291833	Elz
Jhead502	personalfinance	1454349871	Help me get on right p Received promotio
JieShanLanDian	lfg	1454356018	[Online][PBP][All T Zones] Seeking partne
Jimbobmij	jobs	1454351662	Need help dissecting this description to
JimmyTMalice	Pathfinder_RPG	1454353280	Designing a boss fig would appreciate son
JlkesMC	RequestAnyPokemon	1454327347	Post Tagging, Ba Requests and Ribbon Flz
JmodTracker	jmodtracker	15997577464	[Mod Jon] - /u/JagexJon ELI5: How does Jaq

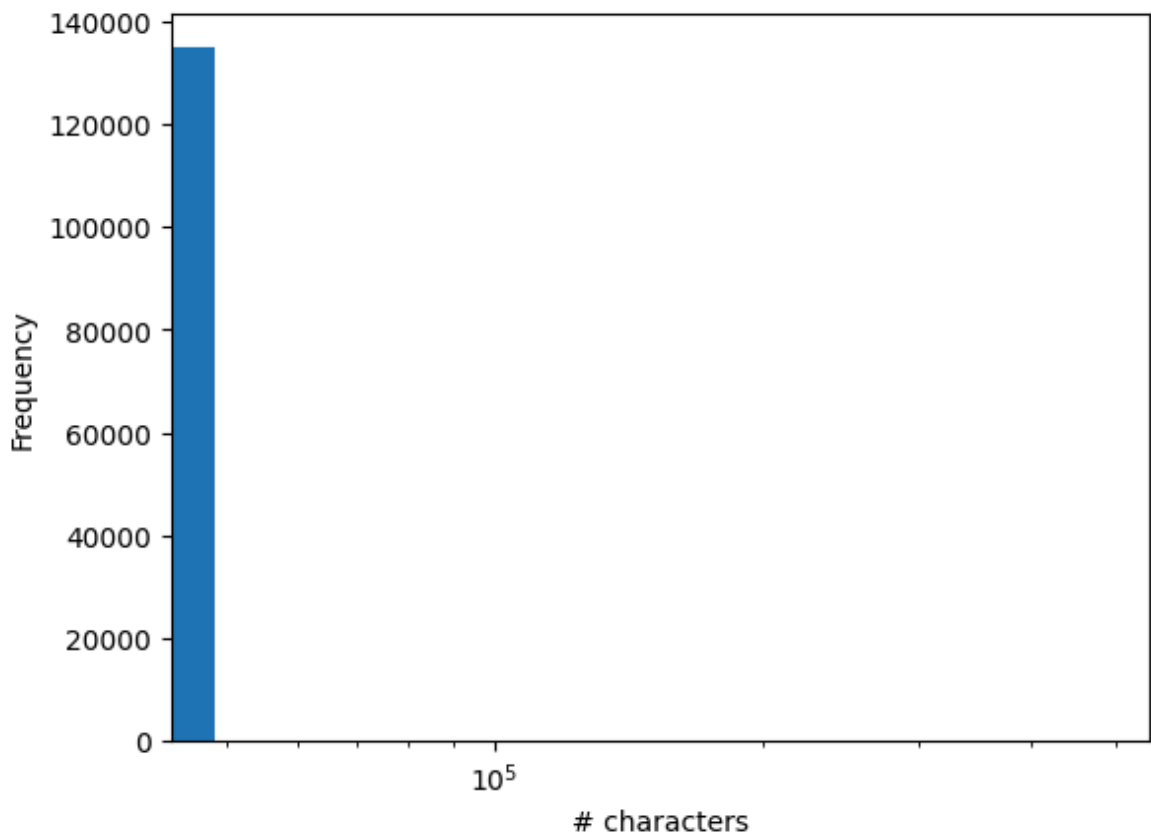
		created_utc	t
author	subreddit		
Joe_Munday	StreetFighter	1454338499	Getting More Bang for Y Knockdown Buck -
Joetruck229	DetroitPistons	1454292544	Opponent Scout Brooklyn Nets. 12.
JoeyPrak736	dirtypenpals	1454289523	19M4A - Celebs and Rul
JohnMcharra	learnprogramming	1454334060	Question about a probab func
JohnShaft	BSA	1454340942	Eagle Court of Ho
JohniiMagii	DnD	1454297462	[3.5e] D&D Drugs Use
Jonathanhong8797	Music	1454318377	WHY I think Rock is los it's popula
Jonch1n	LowSodiumDestiny	1454308405	(suggestion)Another idea making old conte
Jonsnow369	work	1454327846	Why can't bosses just c out, instead of
JoobaTheHutt	AskDocs	1454337082	Pec Proble
JorisBoers	mechmarket	1454330714	[EU-NL][H]Multiple Artis PayPal [W]Trade
Jorlung	EngineeringStudents	1454287277	Aero/Mech Graduate Stuc in the
	gradadmissions	1454287918	Aero/Mech Graduate Stuc in the
JoughPsmyme	dndnext	1454308202	Advice on how to rolepla charismatic Grea
Jreafman	thedivision	2908642831	Shared crafting materia only real complai
JuanFante	relationships	1454314888	Me [25 M] with my first girlfriend [23
Juanabee_	buildapc	1454302916	[Build Help] Upgrad
Jugerknot101	Honda	2908644773	Looking for Civic in Antonio'98 Civic Tr
JuiceBusters	airplanes	1454348275	Seriously, Why Commercial Airpl Inter
Juicy_Gaming	buildapc	1454348569	This is my first gaming build. I wanted t
Jumpinjoe123	FindAUnit	1454293459	[A3][Recruiting]COF wants y

		created_utc	t
author	subreddit		
Junaos	djangolearning	1454349173	[META] The State of /r/djangolearr
Junoncross	techsupport	1454337213	Mysterious process keeping trying to install p
JupiterSaturnNeptune	relationships	1454318546	Me [23 M] with my girlfriend [21 F] broke up
JustACrosshair_	overclocking	1454311452	A catalogue of my overclocking experie
JustJustineHavingFun	AgeplayPenPals	1454302695	[F4F] Chatting about a far fant
Justice1022	TheNarutoWorld	2908660701	Take a WalkAn Unexpected \
Justrayan	askgaybros	1454312110	Almost Strangled (Kil Someone in a Meet
Jvlonden	relationships	1454289493	I [21M] am starting deve feelings for my
Jyg-Lyg	darksouls3	1454299716	Covenant-based "Qu line
K1NG_NAZARA	MLPLounge	1454333781	Hello Plunge, lets discu moral quest
KAQUIK	GlobalOffensiveTrade	1454326450	[STORE] Kara Slaughter M Some Ak sl
KAWAII_SATAN_666	tifu	1454352274	TIFU by trying to poli escape sweat sr
KHolyMatrimony	DarkNetMarkets	1454349083	CanadianForger: High F Cocaine Review (l
KJamesF	gametales	1454320401	[Shadowrun] What we ar the Shadows - Pa
KShizoku	jailbreak	1454343415	[Request] prev PersonalHotspot leechir
KWsonar	GetMotivated	1454295472	[Text] I don't know what "dream" or "goal

Showing only the first 1000 rows × 6 columns

```
In [48]: import matplotlib.pyplot as plt
plt.hist(res.sumChars.to_numpy())
plt.xlabel("# characters")
plt.ylabel("Frequency")
plt.xscale("log")
```

```
/usr/local/lib/python3.12/dist-packages/pyspark/pandas/utils.py:1037: PandasAPIOnSparkAdviceWarning: `to_numpy` loads all data into the driver's memory. It should only be used if the resulting NumPy ndarray is expected to be small.
warnings.warn(message, PandasAPIOnSparkAdviceWarning)
```



```
In [49]: !curl https://2.bp.blogspot.com/-eGskF3n8_Ag/XE7F3P_de2I/AAAAAAAAHU8/WJw0un2nHqM
from wordcloud import WordCloud, ImageColorGenerator
from PIL import Image
```

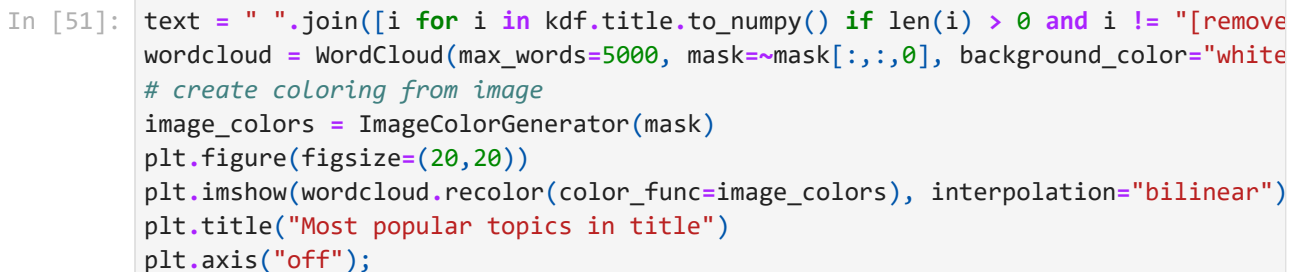
```
mask = np.array(Image.open("reddit.png"))
text = " ".join([i for i in kdf.selftext.to_numpy() if len(i) > 0 and i != "[rem
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
100 76004	100 76004	0 0	146k 0	--:--:--	--:--:--	--:--:--	146k

```
/usr/local/lib/python3.12/dist-packages/pyspark/pandas/utils.py:1037: PandasAPIOnSparkAdviceWarning: `to_numpy` loads all data into the driver's memory. It should only be used if the resulting NumPy ndarray is expected to be small.
warnings.warn(message, PandasAPIOnSparkAdviceWarning)
```

```
In [50]: text = " ".join([i for i in kdf.selftext.to_numpy() if len(i) > 0 and i != "[rem
wordcloud = WordCloud(max_words=5000, mask=~mask[:, :, 0], background_color="white
# create coloring from image
image_colors = ImageColorGenerator(mask)
plt.figure(figsize=(20,20))
plt.imshow(wordcloud.recolor(color_func=image_colors), interpolation="bilinear")
plt.title("Most popular topics in posts")
plt.axis("off");
```

```
/usr/local/lib/python3.12/dist-packages/pyspark/pandas/utils.py:1037: PandasAPIOnSparkAdviceWarning: `to_numpy` loads all data into the driver's memory. It should only be used if the resulting NumPy ndarray is expected to be small.
warnings.warn(message, PandasAPIOnSparkAdviceWarning)
```



[illegible]

Answer: